

Unsupervised Learning: High-dimensional embeddings (t-SNE, UMAP, EVoC) and clustering algorithms applied to chemical tagging of star clusters

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teaching use.

Keywords

chemical tagging, clustering, dimensionality reduction, t-SNE, UMAP, EVoC, stellar abundances, APOGEE, Gaia, unsupervised learning

Abstract

Stars born in the same cluster share a chemical fingerprint, and that fingerprint outlives the cluster: the stars disperse, their element ratios do not. Recovering dispersed siblings from abundances alone (*chemical tagging*) is a clustering problem in a noisy, high-dimensional space, and this workbook is the long-form companion to the IAA-SO 2026 lecture on it. It builds the toolbox one algorithm at a time, each chosen to repair the failure mode of the previous one: K-means, the k -nearest-neighbour primitive, DBSCAN, HDBSCAN*, PLSCAN, t-SNE, UMAP and EVoC, with the mathematics written out, the hyperparameters interpreted, and the assumptions of each method stated rather than hidden. The second half applies the toolbox to 25 open and globular clusters in APOGEE DR19, Gaia DR3 and GALAH DR4, scoring every method against kinematic membership, and reports the result the project actually found: abundances narrow the candidate list, kinematics decide, and a masked spectral autoencoder trained with no abundance labels at all separates clusters better than the ASPCAP abundances do. Every number is reproducible from the accompanying repository, and the failures (including one of our own, withdrawn) are reported next to the successes.

Contents

1.	INTRODUCTION	3
1.1.	A cluster dissolves; its chemistry does not	4
1.2.	The task is clustering, not classification	4
1.3.	Two published verdicts, and the one this workbook adds	5
1.4.	What this workbook measures	5
1.5.	How to read this workbook	6
2.	THE DATA	7
2.1.	APOGEE: abundances for giants	7
2.2.	Gaia and GALAH: the referee and the main sequence	7
2.3.	The abundance space	8
2.4.	Kinematic membership is the referee, not a feature	8
2.5.	The cluster set	9
2.6.	How to reproduce the sample	9
3.	CLUSTERING FUNDAMENTALS	11
3.1.	Four definitions of “cluster”	11
3.2.	Distance in C-space	12
3.3.	Four tasks that are easily confused	13
3.4.	Validation, in outline	13
4.	K-MEANS: THE DEFAULT TOOL	15
4.1.	The objective	15
4.2.	The algorithm, and why it always stops	15
4.3.	Choosing K , and choosing where to start	15
4.4.	What it assumes, and when it breaks	17
4.5.	Is K-means useful for chemical tagging?	17
5.	THE kNN PRIMITIVE	19
5.1.	Neighbours as a classifier	20
5.2.	The k -th neighbour distance is a density estimate	20
5.3.	Cost, and the choice of k	20
5.4.	Our use of it: kNN purity	21
6.	DBSCAN: FOLLOWING THE DENSITY	21
6.1.	Definitions	22
6.2.	The algorithm	22
6.3.	Choosing the two knobs	23
7.	HDBSCAN*: A HIERARCHY OVER EVERY DENSITY	24
7.1.	From points to a density-aware metric	24
7.2.	Building the hierarchy	25
7.3.	What the knobs mean	25
7.4.	Why it fits our problem, and where it still fails	27
8.	PLSCAN: DROPPING THE SIZE FLOOR	27
8.1.	Scale-space, and what persistence measures	28
8.2.	What changes, and what does not	28
8.3.	Where it appears in this workbook	29
9.	VALIDATION: KNOWING WHEN A SCORE LIES	30
9.1.	Internal indices: is the partition tidy?	30
9.2.	External indices: comparison with something the algorithm never saw	31
9.3.	Five ways a score lies	32
9.4.	The protocol this workbook follows	33
10.	t-SNE: A MAP OF NEIGHBOURHOODS	35
10.1.	From distances to probabilities	35
10.2.	The low-dimensional half, and the crowding problem	36

10.3.	What t-SNE does not do.....	37
10.4.	t-SNE on real clusters	37
11.	UMAP: A GRAPH, THEN A LAYOUT	38
11.1.	The high-dimensional graph.....	39
11.2.	The low-dimensional half.....	39
11.3.	Parameters, and why our defaults are what they are	40
11.4.	How it differs from t-SNE in practice	40
12.	EVOC: ONE FIT, THREE STAGES.....	41
12.1.	Stage 1: the graph, built on the metric you meant	41
12.2.	Stage 2: an embedding for clustering, not for looking at	42
12.3.	Stage 3: layers, persistence, and no <code>min_cluster_size</code>	43
12.4.	What EVOC does for us, and what it does not	43
13.	THE BENCHMARK: WHAT WE MEASURED	44
13.1.	The design	44
13.2.	Task 1: separating the clusters from each other.....	45
13.3.	Task 2: finding members in the field	46
13.4.	The curse of dimensionality, measured.....	48
13.5.	Levers, and a lesson about bug fixes.....	48
13.6.	Where this sits in the literature: Casamiquela et al. (2021)	49
13.7.	What this benchmark cannot tell you.....	51
14.	SPECTRA INSTEAD OF ELEMENT RATIOS	52
14.1.	The baseline: PCA of the spectra	52
14.2.	The model: a masked spectral autoencoder	53
14.3.	Why the masking is contiguous	53
14.4.	What it achieves.....	53
14.5.	The error we made, in full.....	55
15.	WHAT MEMBERSHIP IS FOR.....	56
15.1.	The measurement problem	56
15.2.	Distances from red clump giants.....	57
15.3.	The case that decided us: NGC 2243 from two surveys.....	57
15.4.	The finale: better membership, better parameters	58
15.5.	What this does and does not establish	59
16.	THE ASSIGNMENT	60
16.1.	What you are producing	60
16.2.	Setup	60
16.3.	Your cluster	60
16.4.	The three tracks.....	60
16.5.	Levers you may turn.....	62
16.6.	How you are assessed.....	62
17.	CONCLUSIONS	63
17.1.	Eight lessons from the benchmark	63
17.2.	The quiet lesson	64
17.3.	Acknowledgments	66

1. INTRODUCTION

Deck: slides 1–3 (the science question, the data, and the clustering problem) and slide 51 for the reference list.

Every star that exists today was born in a cloud of gas and dust, and most of them were born in the company of thousands of siblings. A cluster forms from one well-mixed reservoir, so its members inherit almost the same chemical composition: element ratios that

Chemical tagging:

Using the element abundances of stars to recover which of them were born together, long after the birth cluster has dispersed.

C-space: The abundance space in which tagging happens: here 16 elements measured from a spectrum, one point per star.

agree to better than 0.1 dex, far below the spread between one cluster and the next. Then the cluster dissolves. Tidal fields, encounters with giant molecular clouds and the shear of the Galactic disc peel the members away one by one, and after a few hundred million years (billions, for the most massive clusters) the stars are scattered across kiloparsecs of the disc, moving on orbits that no longer have anything in common. Their chemistry, however, is frozen. A star keeps the composition it was born with, because nothing in a stellar interior efficiently mixes the surface layers. *Chemical tagging* is the attempt to use that frozen fingerprint to put dispersed siblings back together: to look at a field of stars, all of them currently unrelated, and work out which of them were born in the same lost cluster (Freeman & Bland-Hawthorn 2002; Bland-Hawthorn & Freeman 2016).

This workbook is the long-form companion to the IAA-SO 2026 lecture on that problem. The lecture is a tour of the unsupervised-learning toolbox with one astronomical question attached to it; the workbook goes deeper on both halves: the mathematics of each algorithm, and the measurements we made with it on 25 real open and globular clusters.

1.1. A cluster dissolves; its chemistry does not

The premise is a physical fact about the interstellar medium: mixing is fast and efficient, so a cluster of 10^3 – 10^5 stars forms from gas that was chemically homogeneous to a fraction of a dex. Those stars also formed in one burst, so they share an age. What they do *not* share, once the cluster is gone, is position or velocity. Galactic archaeology therefore faces a reconstruction problem: the cluster is a fossil, and chemistry is the only part of it that is still legible.

Two things make the reconstruction hard, and both are properties of the data rather than of the algorithms.

1. **Precision.** The abundance differences between clusters of similar age and birth radius are small. At 0.05–0.1 dex per element (roughly the precision of a modern survey pipeline) the chemical differences between two coeval solar-metallicity open clusters can be smaller than the measurement noise for the elements in question (Garcia-Dias et al. 2019; Casamiquela et al. 2021). Measured differentially (same instrument, same lines, cluster members only) the internal coherence improves to 0.03 dex, and the cluster-to-cluster differences do not improve with it: the gas the clusters formed from was already mixed to a comparable level, 0.02–0.03 dex of scatter in nearby spiral discs, correlated over scales below 600 pc (Kreckel et al. 2020). The same efficient mixing that makes one cluster chemically coherent is what limits how different two clusters can be.
2. **Contamination.** The field is crowded. In the sample used here, 1 002 stars belong to the 25 target clusters and 357 056 do not: roughly one member in every 300 stars. Any search that returns a group has to be judged on both how many real members it found (recall) and how many of the stars in the group are real members (precision).

1.2. The task is clustering, not classification

Chemical tagging is a *clustering* problem: we are not handing labels to a supervised model, we are asking an unsupervised algorithm to tell us how many groups the data contains and which star belongs to which. That distinction has consequences which reappear throughout the workbook.

WHY IT MATTERS THAT NOTHING IS LABELLED

1. There is no ground truth inside the algorithm. A supervised classifier is scored on held-out labels; a clustering algorithm cannot be, so a clustering result is only as good as the validation you bring to it from outside (**Section 9**).
2. Hyperparameters must be chosen without labels, which is why the method chapters spend as much time on *what the knob means* as on the mathematics that uses it.
3. Feature choice is part of the model. In this problem the same algorithm run on abundances, on a spectral latent, or on kinematics answers different questions, and concatenating two such spaces does not combine their strengths, it dilutes the smaller one (**Section 13**).

1.3. Two published verdicts, and the one this workbook adds

The literature is not unanimous, and the disagreement is instructive because it is not about methods. It is about tasks and precision.

Kos et al. (2017) ran t-SNE on GALAH abundances around the Pleiades and reported success: seven of nine clusters recovered, plus two new Pleiades members six degrees from the cluster centre. Their test case was a young cluster (~ 100 Myr), their verification was kinematic, and their grouping step was visual inspection of the embedding.

Garcia-Dias et al. (2019) asked the same question with APOGEE abundances over 23 clusters and returned a sceptical answer: some clusters are chemically indistinguishable from each other, and abundances alone are enough to identify *families* of clusters rather than individual ones. The reconciliation is not that one study is wrong. The two tagged different objects, at different ages, with different abundance precision, and asked different things of the answer.

A third result belongs on the table too, because it cuts against the pessimistic reading. Hogg et al. (2016) clustered 15 *Cannon* abundances for $\sim 10^5$ APOGEE stars with K-means, using no positional information at all, and showed that abundance-space overdensities really are phase-space clusters. Their abundances reach ~ 0.04 dex, comfortably inside the scatter of the pipeline used here. The scope of any negative result in this field is therefore *these abundances, these clusters*, never “abundances cannot tag”.

The current state of the art sits in between. Spina et al. (2025) trained a graph attention autoencoder on chemistry, orbital similarity and age for $\sim 47\,000$ APOGEE thin-disc stars, obtained 282 groups and recovered five of six open clusters. It is the natural competitor to this work and we have not benchmarked against it; **Section 14** says what we expect a like-for-like comparison to change.

Strong tagging:

Recovering a specific lost birth cluster from chemistry alone: the hard version, and the one this workbook benchmarks.

Weak tagging:

Grouping stars into chemical families. Populations that share a formation epoch rather than a common birth site. The realistic outcome at current precision.

1.4. What this workbook measures

The second half is a benchmark we ran ourselves. We take 25 clusters (18 open, 7 globular) with kinematic membership from Gaia DR3, cut the APOGEE DR19 sample around them, and ask three families of method (t-SNE, UMAP and EVoC) to recover members from 16 abundances, from a 256-dimensional latent of the full spectrum, or from kinematics, all scored against the same kinematic truth. On top of that we train a masked spectral autoencoder that never sees an abundance label and ask whether it tags better than the

abundances themselves.

Four findings organise the results, each developed in its own section:

1. **Abundances narrow; kinematics decide.** Abundance-space clustering separates the 25 clusters only weakly (homogeneity 0.56 for the best projector) where kinematics reach 0.94, and adding abundances to kinematics never helps.
2. **Full-sky collapses; regions work.** Embedding 163 000 stars at once buries every cluster in field stars. The published t-SNE tagging operated on 30–45° regions, and reproduces only there.
3. **The full spectrum beats the element ratios.** A self-supervised latent of 8 575 spectral pixels separates the same stars better than the 16 ASPCAP abundances, and the gap survives removing the globular cluster that dominates the sample. A linear compression (PCA) sits in between, which is the honest way to state the result.
4. **Chemical tagging is a two-survey problem.** APOGEE sees the giants that pin the distance; GALAH sees the main sequence that pins the age. Neither survey alone recovers both cluster parameters.

1.5. How to read this workbook

Deck and workbook are complements. The deck carries the animations (K-means converging, DBSCAN growing, the density sweep, the t-SNE descent, the UMAP layout, the PLSCAN persistence barcode) and the workbook carries the equations, the hyperparameter reasoning and the measured numbers. Where a deck figure moves, the workbook prints a filmstrip of key frames and says so in the caption.



Figure 1

Left: the live deck, where the animated versions of the figures in this workbook run. Right: the repository that produced every number quoted in **Sections 13** and **14**. Both are unlisted but openly reachable.

The chapters follow the order of the lecture. §2 is the data; §3–§12 are the toolbox, each method motivated by the failure of the previous one: K-means (§4), the k -nearest-neighbour primitive (§5), DBSCAN (§6), HDBSCAN* (§7), PLSCAN (§8), validation (§9), t-SNE (§10), UMAP (§11) and EVoC (§12). §13–§15 are the measurements, and §16 is the assignment.

EXERCISES

1. A cluster of 10^4 stars dissolves over a few hundred Myr; a globular cluster of 10^6 stars survives for more than 10 Gyr. Explain from that why chemical tagging should be *easier* for old metal-poor clusters than for young solar-metallicity ones, then check your answer against the per-cluster results in **Section 13**.

2. The field-to-member ratio here is about 300:1. If a search returns 100 candidates for a cluster with 20 true members, what are its precision and recall? Which would you rather sacrifice, and what does that choice cost downstream?
3. Read Hogg et al. (2016) and Casamiquela et al. (2021) side by side: one is optimistic about strong chemical tagging and one is not. Write two sentences each on which *measurement* they disagree about.

2. THE DATA

Deck: slides 2 (the data) and 21 (the scale problem).

Three surveys supply everything this workbook measures. APOGEE provides the chemistry, Gaia the astrometry and radial velocities, and GALAH the southern main sequence that appears in §15. The samples are built from public data releases, and the whole pipeline (download, quality cuts, membership, matrices) is reproducible with three commands (§2.6).

2.1. APOGEE: abundances for giants

The Apache Point Observatory Galactic Evolution Experiment (Majewski et al. 2017) is a near-infrared (H -band, 1.51–1.70 μm) spectroscopic survey of $\sim 10^5$ stars per data release, obtained with the 2.5-m Sloan Foundation Telescope in the north and the du Pont Telescope in the south. The infrared window matters for this project: it sees through the dust that hides the inner disc, and it targets cool giants, which are bright enough to be observed across the whole Galaxy.

Until recently the abundances came from the ASPCAP pipeline. The SDSS-V release used here (DR19 `astraAllStarASPCAP` (Almeida et al. 2023), 1.17 GB, 1 095 480 targets) is produced by its successor, the *Astra* framework. The transfer is not cosmetic; two schema changes are worth knowing about because they bite silently rather than loudly:

- abundances are published as $[X/H]$, not $[X/Fe]$ under a different name, so element ratios must be formed explicitly as $[X/Fe] = [X/H] - [Fe/H]$;
- the quality flags were renamed: `ASPCAPFLAG` became `flag_bad`, `STARFLAG` became `spectrum_flags`.

DR19 also reanalyses rather than replaces DR17: 717 689 of its rows carry `release='dr17'`, all reprocessed by a single version of the pipeline. That detail becomes important in **Section 14**, where an earlier version of our benchmark was nearly derailed by mixing data *products* from the two.

2.2. Gaia and GALAH: the referee and the main sequence

Gaia DR3 (Gaia Collaboration et al. 2023) supplies parallaxes, proper motions and the cross-matched photometry for every star in the sample, all of it carried inside the APOGEE catalogue for convenience. These are the quantities that define membership (§2.4); we never feed them to a tagging method, and **Section 9** explains why that asymmetry is the whole point.

GALAH DR4 (De Silva et al. 2015; Buder et al. 2021) covers declination $\lesssim +25^\circ$ with a high-resolution optical spectrograph, which means it sees the *main sequence* where

ASPCAP / Astra:

The abundance pipelines: ASPCAP up to SDSS-IV DR17, Astra from SDSS-V DR19 onwards. Same philosophy (fit synthetic spectra to the observation) different catalogues and column names.

Abundance

precision: The per-element statistical error on $[X/H]$, typically 0.02–0.1 dex for APOGEE giants, depending on the element and the signal-to-noise ratio. It sets the ceiling on what chemistry can tag.

APOGEE sees giants. It enters only in §15, where the two surveys are combined to recover a cluster age and distance at the same time.

2.3. The abundance space

Everything is standardised into one matrix: 16 numbers per star, one column per element, built from the survey catalogue by a fixed sequence of steps. Each step is a modelling choice, not housekeeping, and each has a failure mode we have hit.

1. **Quality cuts.** Signal-to-noise ratio ≥ 100 , ASPCAP flag zero, stellar flag zero. Cuts this aggressive remove the noisy tail at the cost of sample size; loosening them changes the answer (§13, the levers).
2. **The element set.** C, N, O, Na, Mg, Al, Si, S, K, Ca, Ti, V, Cr, Mn, Ni (the α and iron-peak elements) plus [Fe/H] as the sixteenth coordinate. This is the set used by the reference papers, so results are comparable with them.
3. **Missing values.** Weak lines go undetected in metal-poor stars and their abundances come back as NaN. Dropping every star with a NaN deletes the metal-poor globulars *entirely*: M 15 and M 92 vanish from the sample, and with them the cleanest tagging cases in the whole project. We therefore keep stars with at least 8 of 16 finite elements and fill the remaining gaps with the column median. This single choice is the difference between a sample that contains the interesting objects and one that does not.
4. **Standardisation.** Each element is shifted to zero median and scaled to unit standard deviation, so that no element dominates the distance metric by virtue of its units. (An early bug in this project applied inverse-variance weights *before* this step, which silently cancelled them; see §13.)
5. **Row normalisation.** Each star’s 16-vector is scaled to unit L_2 norm. The reason is metric compatibility: after this step, Euclidean distance is a monotone function of cosine distance, so t-SNE and UMAP (which use Euclidean) and EVoC (whose native metric is cosine) all see the same geometry, and the comparison between them is fair. It is also the single most effective precision lever in the pipeline: without it, density-based clustering merges the whole field into one blob.

2.4. Kinematic membership is the referee, not a feature

To score anything we need labels, and this is where chemical tagging is unusually lucky: a cluster is also a kinematic object. Its members currently share a position on the sky, a parallax, a proper motion and a radial velocity, because they were born at the same place with almost the same velocity. We use those four observables, σ -clipped around the cluster centre, to label each star *member* or *field*.

The circularity risk is obvious: if kinematics are used to build the labels, then a method that receives kinematics as an input is being scored on its own input. We therefore treat kinematics as a *ceiling*, not a competing signal. The number that says how well *any* chemistry-based method could possibly do on this sample. Published cluster catalogues provide an independent cross-check: scoring against Simbad memberships gives numbers within a few percent of the kinematic ones, and both are reported in the benchmark (Dias et al. 2002; Harris 1996).

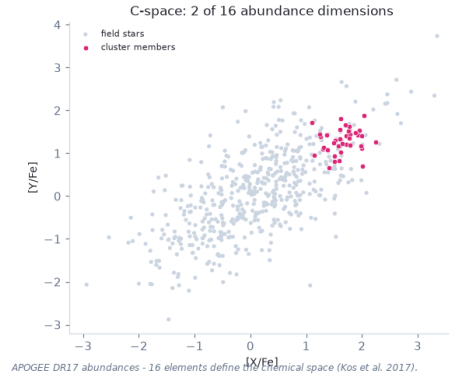


Figure 2

The shape of the problem, in two of the sixteen dimensions (from the deck). Field stars in grey; the members of one cluster in red. The cluster is a tight, slightly elongated clump embedded in a cloud several orders of magnitude larger: a small overdensity in a big, structured background. Everything in §3–§12 is a strategy for finding objects like this one.

2.5. The cluster set

The target list is the 23 clusters of Garcia-Dias et al. (2019) plus the Pleiades of Kos et al. (2017), and, for the two-survey work, the southern open clusters NGC 2243 and Collinder 261. Twenty-five objects in total: seven globular and eighteen open. They span two orders of magnitude in age, from the Pleiades at ~ 100 Myr to NGC 188 at 7.6 Gyr, and two orders of magnitude in mass. The counts below come from the full DR19 run (357 056 field stars). They are *row counts*, not star counts: the sample contains repeat entries for the same star, a defect quantified in §9 and one we flag here rather than at the end because it affects how every macro average in this workbook should be read.

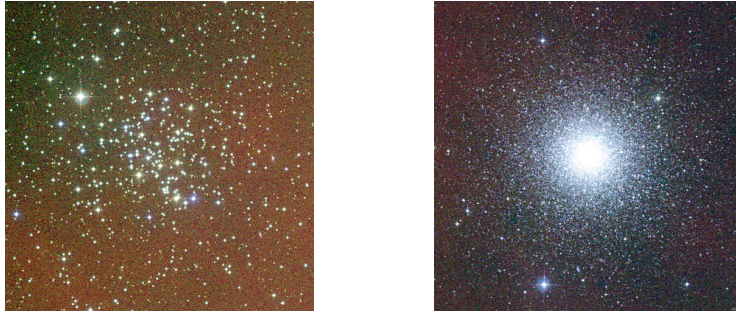


Figure 3

Two of the targets, in the optical (DSS2 colour composites centred on the cluster): M 67 (left), an old solar-metallicity open cluster, and M 3 (right), a metal-poor globular. The two objects illustrate the range of the sample and, in the results, the range of difficulty: the globular is chemically distinct from the field around it, the open cluster is not.

2.6. How to reproduce the sample

The pipeline is one package with a single configuration file, so every number in this workbook can be regenerated from a fresh checkout. It is written on `numpy` (Harris et al. 2020), `pandas`

Table 1 The 25 target clusters and the number of kinematic members each contributes in the DR19 sample. Globulars are marked with a †. The asymmetry in the counts is real and consequential: M 67 alone supplies 23% of all members, while three clusters supply fewer than ten.

Cluster	n_{mem}	Cluster	n_{mem}
Berkeley 17	7	M 3 [†]	154
Berkeley 66	12	M 5 [†]	67
Berkeley 71	14	M 13 [†]	34
Collinder 261	22	M 15 [†]	31
IC 166	35	M 71 [†]	39
King 5	12	M 92 [†]	25
King 7	8	NGC 1245	21
M 67	230	NGC 1798	13
Pleiades	23	NGC 188	25
		NGC 2158	6
		NGC 2243	36
		NGC 2420	19
		NGC 6791	45
		NGC 6819	62
		NGC 7789	47
M 107 [†] contributes 15 members. Total: 1002 members in 25 clusters.			

Counts from the full-sky DR19 run (cluster run), field star cap disabled, on the member frame the DR19 baseline scores: every member row with at least nine finite abundances out of sixteen. Reproduce with `python scripts/population_counts.py -baseline-min-finite 9`; the counts are identical for any floor between nine and fifteen. Member counts differ slightly between runs because the clustering must first find a group before its overlap with the truth can be measured; these are the truth-side counts.

(McKinney 2010), `scikit-learn` (Pedregosa et al. 2011), `astropy` (Astropy Collaboration et al. 2022) and `matplotlib` (Hunter 2007), with the clusterers taken from their reference implementations. The commands below are all you need to rebuild the sample and the benchmark.

HOW WE RAN IT

```
git clone github.com/garciadias/iaa-advanced-neural-networks-2026-draft
cd iaa-advanced-neural-networks-2026-draft
uv sync # Python 3.13 environment
uv run cluster download # 1.17 GB, SDSS-V DR19 Astra ASPCAP
uv run cluster run --fast # ~1 min on a laptop, capped field
CLUSTER_FAST=0 uv run cluster run # full all-sky run, 10-20 min
```

Every flag lives in `src/cluster/config.py` and every flag also has an environment override with the `CLUSTER_` prefix, which is how the variants in §13 were produced. The Docker path (`docs/docker.md`) exists for machines without `uv`.

setting	value used throughout
SNR_MIN	100
dwarf cut (DWARF_ONLY)	off
elements	16: [C/Fe] ... [Ni/Fe] (15 ratios) + [Fe/H]
imputation	median, requiring ≥ 8 finite elements
standardisation	on (zero median, unit s.d.)
row normalisation	on
element weights	off
HDBSCAN.min_cluster_size	5
TSNE.perplexity / UMAP.n_neighbors	30 / 15
random_state	42

Table 2 The pipeline’s settings for every number in this workbook, unless a variant is named. Overrides are environment variables with the CLUSTER_ prefix. CLUSTER_NORMALIZE_ROWS=0 is the one that most changes the answer.

EXERCISES

1. Take the 16-element vector for one star in the sample. Standardise it and then row-normalise it. Show algebraically that after row normalisation the Euclidean distance between two stars is a strictly monotone function of the cosine distance between them. Why does EVoC care?
2. MIN_FINITE_ELEMENTS defaults to 8 of 16. Set it to 16 (strict complete-case) and count how many of the 25 clusters survive with at least five members. Which clusters are lost, and what does that do to any conclusion about the surviving ones?
3. M 67 contributes 230 of 1002 members. Recompute a macro-average metric over clusters with and without M 67 in the pool. How much of the published summary depends on one cluster?

3. CLUSTERING FUNDAMENTALS

Deck: slides 3 (the clustering problem), 18 (no free lunch) and 19–20 (validation).

Before any algorithm, two questions deserve a straight answer: what do we mean by “a cluster”, and what does a distance in 16 dimensions actually measure? The answers are not academic. Most of the failures in **Section 13** are failures of an assumption about one of those two things, not failures of an implementation.

3.1. Four definitions of “cluster”

A clustering algorithm does not discover the clusters that are in the data; it discovers the clusters that its own definition of “cluster” implies. The definitions in play in this workbook are:

- **Centroid.** A cluster is a set of points closer to one centre than to any other. This is K-means (§4), and it implies convex, roughly equal-sized groups.
- **Density.** A cluster is a connected region where the local point density exceeds a threshold. This is DBSCAN and its descendants (§6–§8), and it implies that clusters

Multimodality: A distribution with more than one peak. Clustering is only meaningful if the peaks are physical: here, if stars of different birth sites occupy different regions of C-space.

No free lunch: No algorithm is best on every problem: each encodes assumptions about shape, scale, density and metric. Choosing a method is choosing which assumptions are safe for your data.

and noise are distinguishable by density alone.

- **Connectivity.** A cluster is a connected component of a graph built from nearest neighbours, with edges weighted by similarity. This is the graph view that t-SNE, UMAP and EVoC share (§10–§12).
- **Statistical.** A cluster is a component of a mixture model whose parameters are inferred by maximum likelihood (the EM algorithm, Dempster, Laird & Rubin 1977). The cluster boundaries are then soft: a star has a probability of belonging to each group.

The reference chapter behind the lecture’s framing (Garcia-Dias et al. (2020), a methods chapter written for a psychiatry audience) makes the same point from the other direction: clustering is worth attempting when the joint distribution of the variables is *multipeaked*, and the first decision is whether the data are multimodal at all. If a distribution is unimodal, every algorithm will return groups anyway, and every one of them will be an artefact.

3.2. Distance in C-space

Every algorithm in this workbook reduces to a similarity judgement, and all of them use either Euclidean or cosine distance. Two transformations in **Section 2** make those two compatible, and one transformation is a genuine modelling choice:

$$d_{\text{Euclid}}(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2, \quad d_{\text{cos}}(\mathbf{x}, \mathbf{y}) = 1 - \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2}.$$

After per-element standardisation and per-star L_2 normalisation, all stars lie on the unit sphere in 16 dimensions and d_{Euclid} is a monotone function of d_{cos} , so t-SNE and UMAP (Euclidean internally) and EVoC (cosine) are looking at the same geometry, and comparing them is fair. The Euclidean alternative without normalisation is not wrong, but it inherits a scale from the standardisation: elements with a long tail of outliers become the elements that decide every distance.

High dimension does the rest. Two effects matter here.

1. **Concentration.** In d dimensions, the ratio between the furthest and nearest neighbour distance shrinks as d grows: distances *concentrate* around a common value. A density threshold that worked in 2-D has to be re-tuned, and the contrast between “near” and “far” degrades. This is the everyday experience of the curse of dimensionality in clustering: the kNN graph of 16-D abundances has edges that are only slightly shorter than the edges you would get by chance.
2. **Dilution by concatenation.** Adding dimensions that carry no cluster information can only reduce the relative contribution of the dimensions that do. The consequence in this project is concrete. On the same 982 member stars, in the same pipeline (**Section 13**), the cluster-only homogeneity is 0.56 from 16 abundances, 0.94 from 4 kinematic coordinates, and 0.75 from the 20-dimensional concatenation of the two. The blend is *worse* than the kinematics alone: the chemistry did not add information, it added dimensions.

That last result deserves emphasis because the intuitive fix (“if both signals are useful, feed both”) is the one most people try first:

THE TRAP OF CONCATENATION

Fusing two signals of very different dimensionality into one Euclidean space is not a fusion of information; it is a weighted sum in which the weight of each channel is set by how many coordinates it happens to contribute. A 256-dimensional spectral latent next to 4 kinematic coordinates gives the spectrum 98% of the vote, whatever the physics says. The two ways out are to reduce the high-dimensional channel before concatenating it, or to combine the two at the level of the *decisions* rather than the features: the two-stage pipeline of §13, which uses kinematics for recall and a spectral distance for precision.

3.3. Four tasks that are easily confused

The literature (and our own first drafts) mixes up four different questions that all get called “clustering”. Keeping them separate is the single most useful discipline in this project.

1. **Membership determination.** For a given cluster, label each star in a field *member* or *field*. This is the operationally useful task, and it is what a survey pipeline actually needs.
2. **Discovery.** Blind search: find groups without knowing in advance where to look or what they should look like. This is what chemical tagging means in the “strong” sense of §1, and it is much harder: the algorithm must invent the groups as well as find their members.
3. **Refinement.** Clean an existing candidate list. Remove field contaminants from a known cluster’s sample. This is the task our two-stage pipeline performs, and it is measurably easier than discovery.
4. **Cluster-only separation.** Take the members of several clusters, with no field stars present, and ask whether the clusters are distinguishable from one another. This is the experiment of Garcia-Dias et al. (2019), recreated in §13 as the *baseline*. Its score is an upper bound on what any field-level search could do: if clusters cannot be told apart when nothing else is present, they cannot be recovered from a field either.

3.4. Validation, in outline

Because there are no labels, validation is a design decision, not a calculation. The toolbox has two halves, developed in **Section 9**:

- **Internal** indices (sum of squared errors, silhouette, the dip test) measure how *tidy* a partition is. They are computed from the data alone, they are cheap, and they cannot tell you whether the partition is real.
- **External** indices compare the partition with something the algorithm never saw. In this project that something is kinematic membership; in general it is any independent measurement.

The dangerous middle is a number that looks external but is not: a metric that uses the quantity you are trying to predict, or a baseline that has collapsed to two groups and scores well for the wrong reason. Both appear in our own results and are flagged there.

Table 3 The assumptions each method in this workbook makes explicit. Read it as a checklist: when a method fails on our data, one of these entries is usually the reason, and the next chapter takes over because of it.

Method	Assumes	Fails when
K-means	clusters are convex, similar in size, separable by a centre; K known	shapes are elongated or nested; outliers pull centres
k NN	a metric is meaningful and local density is informative	dimensionality is high; the k is chosen without looking at the data
DBSCAN	one density threshold separates cluster from noise	clusters and noise share a density; densities vary between clusters
HDBSCAN*	density varies, but a single minimum cluster size applies	clusters are smaller than the floor; the metric is misleading
PLSCAN	persistence, not size, identifies a cluster	clusters are genuinely short-lived in scale-space (rare)
t-SNE	local neighbourhoods are what matter	inter-cluster distances are being interpreted
UMAP	the manifold is locally uniform	global geometry is treated as metric information
EVoC	a cosine-style graph and one internal embedding serve all scales	the metric is wrong for the data

EXERCISES

1. The concentration of distances can be demonstrated in five lines: sample $n = 1000$ points uniformly in $\mathbb{R}^d$, compute for each point the ratio of the distance to its nearest and to its furthest neighbour, and plot the mean ratio against d for $d = 1, \dots, 50$. At what d does the nearest neighbour stop being meaningfully nearer than the furthest one?
2. Standardise the 16-D matrix, then row-normalise it. A star with an unusually low [Fe/H] and a star with an unusually high one now have the same norm. What did normalisation do to the information about overall metallicity, and why is that mostly a feature here rather than a bug?
3. Pick three clusters from **Table 1** (one with more than 60 members, one with about 20, one with fewer than 10) and write down what “ K is known” would even mean for a field of 25 clusters with these counts. Which of the four tasks of §3.3 does a fixed K violate?

4. K-MEANS: THE DEFAULT TOOL

Deck: slides 4–11: the K-means tour, ending on its limits.

K-means is where every clustering analysis starts, and it is worth understanding precisely, because its failure modes are the ones the rest of the toolbox was invented to repair. The method was proposed by MacQueen (1967) and its modern formulation (the one everyone implements) is due to Lloyd (1982).

4.1. The objective

Given n points $\mathbf{x}_1, \dots, \mathbf{x}_n$ in $\mathbb{R}^d$ and a fixed number of groups K , K-means looks for K centres $\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K$ and an assignment of every point to one centre that minimises the *sum of squared errors* (SSE), also called the within-cluster sum of squares:

$$J = \sum_{k=1}^K \sum_{\mathbf{x} \in C_k} \|\mathbf{x} - \boldsymbol{\mu}_k\|_2^2, \quad \boldsymbol{\mu}_k = \frac{1}{|C_k|} \sum_{\mathbf{x} \in C_k} \mathbf{x}. \quad (1)$$

The second half of the equation is not a separate model choice but a consequence of the first: for a fixed assignment, the centre that minimises the sum of squared distances to the members is the arithmetic mean. Two properties follow immediately and both are assumptions in disguise.

- Because J penalises distance *squared*, the objective is dominated by the points furthest from their centre. A single outlier can drag a centre across the field. K-means is not robust; medians (K-medians) or medoids are the standard remedy when outliers are expected.
- Because every point is assigned to its nearest centre, the partition is a *Voronoi tessellation*: cells are convex and their boundaries are hyperplanes. A cluster with a crescent or dumbbell shape cannot be represented, however obvious it looks to the eye (**Figure 7**).

SSE / inertia: The sum of squared distances from each point to its assigned centre: the quantity K-means minimises. Also called inertia in scikit-learn and within-cluster sum of squares (WCSS).

Voronoi cell: The region of space closer to one centre than to any other. K-means partitions the plane into Voronoi cells by construction.

4.2. The algorithm, and why it always stops

Lloyd’s algorithm is a two-move loop. *Assign* every point to its nearest centre; *update* every centre to the mean of its members; repeat. Each move can only reduce J : the assignment step is optimal for fixed centres, and the update step is optimal for fixed assignment. J is bounded below by zero and takes finitely many values (there are finitely many partitions of n points into K groups), so the loop must terminate, in practice after a few dozen iterations. It terminates at a *local* minimum, and the distinction matters: K-means is a coordinate-descent algorithm on a non-convex objective, so different starting centres end in different answers.

4.3. Choosing K , and choosing where to start

K is an input, which is uncomfortable in a problem where the number of clusters is exactly what we are trying to learn. Three standard answers, and a warning:

1. **Elbow.** Plot J against K ; J decreases monotonically (more centres always fit better), and the curve is expected to bend where extra groups stop buying real structure. It

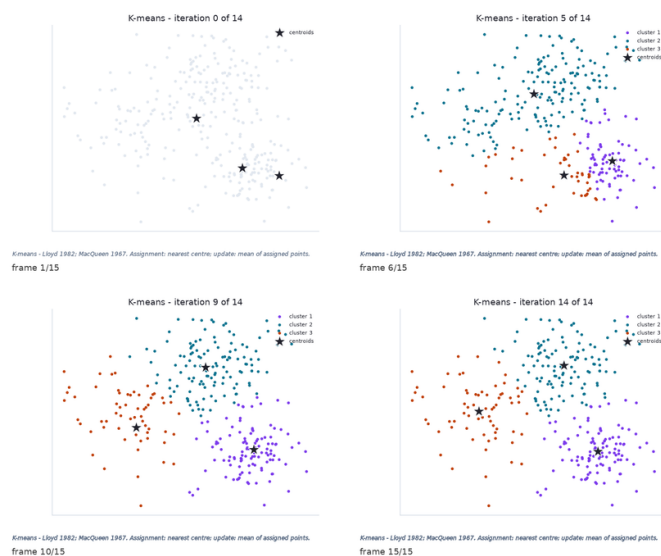


Figure 4

The K-means loop on a synthetic 2-D dataset, sampled at four points in the animation: initial centres, first assignment, first update, converged. The workbook prints the key frames; the deck plays the full sequence (**Figure 1**).



Figure 5

The same four states, in the deck's step figures. Stars are the centres, dots the data: (a) all points unassigned with K centres dropped at random; (b) assignment to the nearest centre; (c) centres moved to the mean of their members; (d) the fixed point of the iteration, no point changes hands, so the loop stops.

is a heuristic: on smooth data there is no elbow, and on clustered data with a broad size distribution the bend is shallow.

2. **Silhouette.** For each point, compare its mean distance to its own cluster with its mean distance to the nearest other cluster; average over points and pick the K that maximises the score (Rousseeuw 1987). It is a ratio of within- to between-cluster

distance, so it also penalises non-convex clusters.

3. **Model selection.** Compare K by held-out likelihood under a Gaussian mixture fitted by EM (Dempster, Laird & Rubin 1977). K-means is the hard assignment limit of that model with spherical, equal-variance components, which is precisely the assumption list of **Table 3**.

The warning is that in this project we *cheat*, deliberately. The catalogue tells us there are 25 clusters, so the benchmarking arms are handed the true number of groups. That makes the comparison between methods clean, and it flatters all of them relative to a blind search, which is why §13 reports field retrieval (a discovery-like task) and cluster-only separation (a K -known task) separately.

Initialisation has the same status: a modelling decision that changes the answer. The classical approach is to run the loop from several random starts and keep the best J ; the modern default is *k-means++* (Arthur & Vassilvitskii 2007), which samples the first centre uniformly and each subsequent centre with probability proportional to $D(\mathbf{x})^2$, the squared distance to the nearest centre already chosen. The effect is to spread the initial centres out, which typically reaches a better local optimum in fewer restarts.

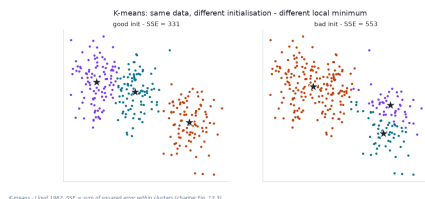


Figure 6

Initialisation sensitivity (from the deck): the same data, two random starts, two different final partitions, and one of them splits a real group. Nothing about the algorithm changed; only the seed did. In a clustering paper this is the first thing a referee will ask about, and it is why every number in this workbook is reported as a mean over seeds (seven, in the head-to-head of §14).

4.4. What it assumes, and when it breaks

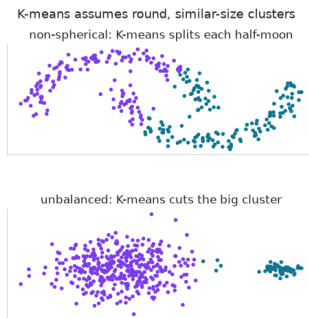
THE ASSUMPTION LIST

K-means assumes that clusters are (i) separable by a centre, (ii) roughly convex, (iii) of comparable size and variance, (iv) all present in equal proportion, and (v) that K is known. It additionally assumes that the metric is meaningful: that the dimensions have been standardised, or that the scales carry physical meaning. When it fails, it fails in a characteristic way: it splits large or elongated groups and merges neighbouring small ones, and it never returns “noise”.

4.5. Is K-means useful for chemical tagging?

Yes: with the right expectations, and in the literature it has been used twice in ways that matter here.

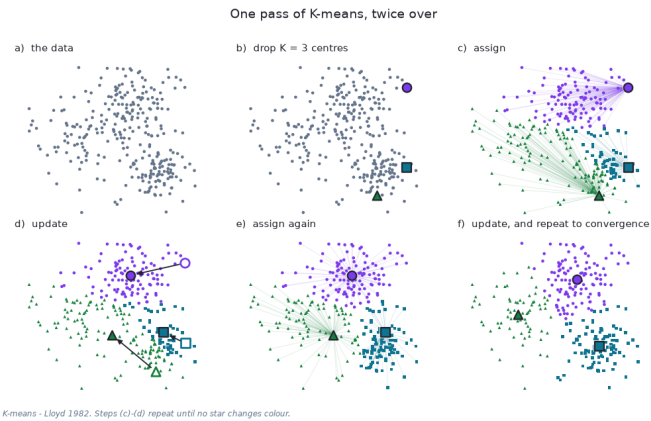
Hogg et al. (2016) clustered $\sim 10^5$ APOGEE stars in a 15-dimensional abundance space with K-means, with no positional information, and recovered groups that are genuinely phase-space structures. That result defines what abundance-space overdensities can be at



K-means - Lloyd 1982. Failure modes from the Clustering analysis chapter (Fig. 13.4).

Figure 7

K-means on two elongated, interleaved structures. The natural groups are obvious to the eye and unavailable to a partition by nearest centre: the objective can only be reduced by cutting across the arms. This is the failure that motivates every method in the rest of the workbook.



K-means - Lloyd 1982. Steps (c)-(f) repeat until no star changes colour.

Figure 8

The full K-means story in one grid, as used in the lecture: the loop, an initialisation that goes wrong, the failure on non-convex shapes, and the diagnostic that the objective can be lowered without the answer getting better.

~ 0.04 dex precision. And Garcia-Dias et al. (2018) applied K-means to 153 847 normalised APOGEE spectra for spectral classification rather than for tagging, and the classes it returns do track the stellar parameters. Dwarfs separate from giants, bulge from halo. But the same paper's conclusion is the opposite of tidy: flux space has no obvious groups, a discrete classification does not organise parameter space neatly, and the solution moves with the initialisation. That is the same warning as the row-order result of §9, in a different space. K-means is useful when the groups are blobs; the spectra are not blobs, and the abundance space of §13 is only partly one.

In **Section 13** the same algorithm is run on the same member matrix as the other arms (the 1002-row matrix of **Table 1** in this chapter's summary box, and the 982-star deduplicated population of **Table 4** and **Table 10**) so that the comparison between “the classical tool” and the newer ones is measured rather than asserted.

K-MEANS ON OUR MEMBER MATRIX

`article/scripts/kmeans_baseline.py` loads the DR19 sample, takes the same member-only matrix the other benchmark arms use (1002 stars, 16 abundances, 25 clusters, ≥ 5 members), runs K-means with $n_{\text{init}} = 10$ and reports the paper’s merit functions alongside the SSE curve. The measured result is worth stating plainly, because it is not the one the gospel predicts: at $K = 25$, K-means reaches **homogeneity 0.449** (completeness 0.493, v-measure 0.470, accuracy 0.296) (higher than t-SNE’s 0.292 and EVoC’s 0.409 on the same stars, and within reach of UMAP’s 0.547. Two caveats travel with it. At $K = 2$ the same matrix gives homogeneity 0.228 with completeness 0.954: the “families” reading, in which almost everything lands in one group. And raising K to 40 lifts homogeneity to 0.505 while completeness falls to 0.474) over-splitting buys homogeneity, which is why the paper’s metric pair must always be quoted together.

EXERCISES

1. Show that the assignment and update steps each cannot increase J , and conclude that the algorithm terminates. Then construct a one-dimensional example (three points, two centres) where the global optimum and Lloyd’s fixed point differ, and draw the two partitions.
2. Implement k-means++ seeding in ten lines of NumPy and compare the final J over 50 random datasets against a uniform-random initialisation. How much of the improvement comes from the seeding rule, and how much from the multiple restarts?
3. K-means with $K = 2$ on our member matrix asks a different question from $K = 25$: not “which cluster is this star in” but “which family”. Run both and compare the homogeneity. Which of the two numbers would you quote to support strong chemical tagging, and why is that the wrong number?
4. Replace the Euclidean distance in the assignment step with cosine distance, keeping the mean update. What happens to the objective’s guarantee of monotone decrease, and does the algorithm still terminate?

5. THE KNN PRIMITIVE

Deck: slide 12, with slide 23 for the core distance $\kappa(x)$.

Before density-based clustering and before graph embeddings, there is one operation that all of them depend on: find a point’s k nearest neighbours. It is the oldest idea in the toolbox (Cover & Hart 1967), it is simple enough to be taken for granted, and in this project it turns out to be both a clustering ingredient and a validation metric in its own right.

Core distance:

$\kappa_k(\mathbf{x})$, the distance from a point to its k -th nearest neighbour. Large κ means an isolated point; small κ means a dense neighbourhood.

kNN graph: The directed graph in which every point is joined to its k nearest neighbours. Its nodes are stars, its edges similarity, and it is the substrate that HDBSCAN*, UMAP and EVoC all build on.

5.1. Neighbours as a classifier

The classical k -nearest-neighbour rule is a supervised method: to classify a point, find its k nearest labelled neighbours and take a vote. Its appeal here is diagnostic rather than predictive. A k NN vote hides no assumptions about the shape of a decision boundary, so an error rate measured with k NN approximates the best you can do with a local rule in that space. When we ask whether 16 abundances carry cluster information at all (§13), a nearest-neighbour view answers the question directly: do members share their neighbourhood with other members?

5.2. The k -th neighbour distance is a density estimate

Distances and densities are two descriptions of the same local geometry. The volume of the ball containing exactly a fraction k/n of the data is $V \propto \kappa_k(\mathbf{x})^d$ in d dimensions, so a local density estimate follows from the measured distance alone:

$$\hat{\rho}(\mathbf{x}) \propto \frac{k}{n \kappa_k(\mathbf{x})^d}. \quad (2)$$

Two consequences are used repeatedly in the chapters that follow. First, κ converts a distance into a density, which is the operation DBSCAN performs with its global ε and HDBSCAN* performs locally. Second, the conversion involves d , the dimension: in 16 dimensions a factor of two in density is a mere 4% change in κ , which is why density-based methods on abundance data are so sensitive to the metric and to the normalisation chosen in §2.

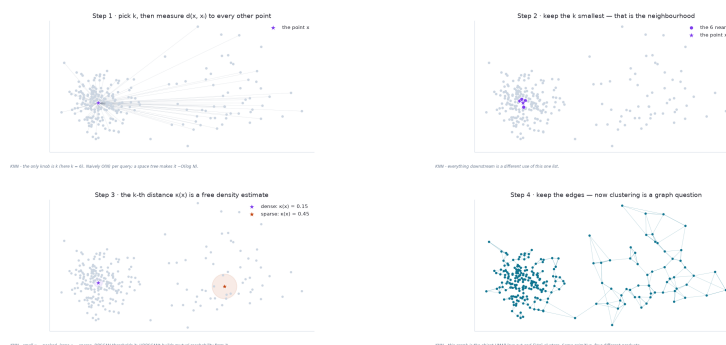


Figure 9

The primitive, in four steps (deck figures): (a) measure every pairwise distance from the query point (black); (b) keep the k smallest. The neighbourhood; (c) the k -th distance is the core distance κ_k , a density estimate read off the data; (d) keep the edges and you have a graph, which is all the rest of the toolbox needs.

5.3. Cost, and the choice of k

Exact k NN costs $O(n^2d)$ for n points in d dimensions when computed brute force, which for 3.6×10^5 field stars and $d = 16$ is about 10^{12} distance evaluations: feasible on a cluster, painful on a laptop. The practical answers are spatial indices (KD-trees, ball trees, which degrade badly above roughly 20 dimensions) and approximate graphs (NN-descent, the algorithm UMAP uses, which is what makes it fast). This distinction matters when reading

performance claims: “UMAP is fast” is a statement about its graph construction, not about exact k NN.

The choice of k is the usual bias–variance trade: small k gives a high-variance density estimate that fragments clusters; large k smooths away small ones. Our pipeline uses $k = 15$ for both the UMAP graph and EVoC’s internal graph, and $k = 10$ for the purity score below. All three are conventional defaults, and all three are knobs a careful user should vary to see whether the conclusion moves. **Section 13** reports one such variation: UMAP at $n_{\text{neighbours}} = 5$ performs measurably worse than at the default 15.

5.4. Our use of it: k NN purity

The one place where k NN enters our results directly is validation. For each star that is a known member of a cluster, count the fraction of its 10 nearest neighbours in the embedding that belong to the same cluster; average over members. The result (k NN *purity*) is a cohesion score with no hyperparameters beyond k , no clustering step, and no way to tune yourself a good answer:

$$\text{purity} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \frac{|\{j \in \text{kNN}_{10}(i) : \text{cluster}(j) = \text{cluster}(i)\}|}{10}, \quad (3)$$

where $\mathcal{M}$ is the set of known members. It is the automated version of the manual test in Kos et al. (2017) (draw a polygon around the visual group in the embedding and see whether the members are inside it) and it is reported for every method in §13. A purity near 1 means the members of a cluster genuinely occupy a neighbourhood of the space; a purity near the field fraction means the embedding knows nothing about them.

EXERCISES

1. Derive the density estimate of §5.2 from the volume of a ball containing k points. Then compute, for the DR19 abundance matrix, the spread of κ_{15} across members and field stars. Is the cluster signal visible in the core distance alone, before any clustering?
2. Build the 15-NN graph of the member matrix and compute the purity of the known members in the *raw* C-space. Compare with the purity after t-SNE, UMAP and EVoC. Which method moves the number most, and does it move it by separating members or by compressing the field?
3. ANN libraries advertise large speedups over exact k NN. Measure the disagreement rate between an approximate graph (UMAP’s NN-descent) and an exact one on a random 5000-star subsample of the DR19 matrix. Is the approximation good enough that the clustering result is unchanged?

6. DBSCAN: FOLLOWING THE DENSITY

Deck: slides 13–17, with slide 21 for the single-threshold problem.

K-means asks how far a star is from a centre. DBSCAN (density-based spatial clustering of applications with noise (Ester et al. 1996)) asks a different question: how many neighbours does a star have within a fixed radius? That single change buys three things K-means cannot

offer: clusters of arbitrary shape, an explicit notion of noise, and no need to know K in advance.

ε (**epsilon**): The neighbourhood radius. It sets the density threshold: a point is dense if it has minPts neighbours within ε of itself.

minPts: Minimum number of points (including the point itself) required inside the ε -ball for a point to be a core point. Typical defaults are $d + 1$ for d dimensions, or $\ln n$.

6.1. Definitions

Fix two parameters: a radius ε and a minimum count minPts . Let $N_\varepsilon(\mathbf{x})$ be the set of points within distance ε of $\mathbf{x}$ (its ε -neighbourhood).

- **Core point.** $\mathbf{x}$ is a core point if $|N_\varepsilon(\mathbf{x})| \geq \text{minPts}$. Only core points can extend a cluster.
- **Border point.** A non-core point that lies inside the ε -neighbourhood of a core point. It joins that cluster, but cannot recruit further points.
- **Noise.** Everything else: not core, and not within ε of a core point. It is labelled -1 and never counted as a member of anything.
- **Directly density-reachable.** $\mathbf{y}$ is directly density-reachable from $\mathbf{x}$ if $\mathbf{y} \in N_\varepsilon(\mathbf{x})$ and $\mathbf{x}$ is a core point. Note the asymmetry: a border point is reachable *from* its core neighbour but not the other way round.
- **Density-connected.** Two points are density-connected if there is a chain of core points, each directly reachable from the previous, touching both. Density-connectivity is symmetric, and a DBSCAN cluster is a maximal set of density-connected points.



Figure 10

DBSCAN growing a cluster, sampled at four points in the animation: core points are identified, a seed point starts a cluster, connected core points chain it along the crescent, and border points attach to the chain. The points that never enter a core's ball stay noise, permanently.

6.2. The algorithm

The plain version is short enough to write out. For each point not yet visited: if it is not a core point, mark it noise for now and move on; if it is, start a new cluster and grow it by

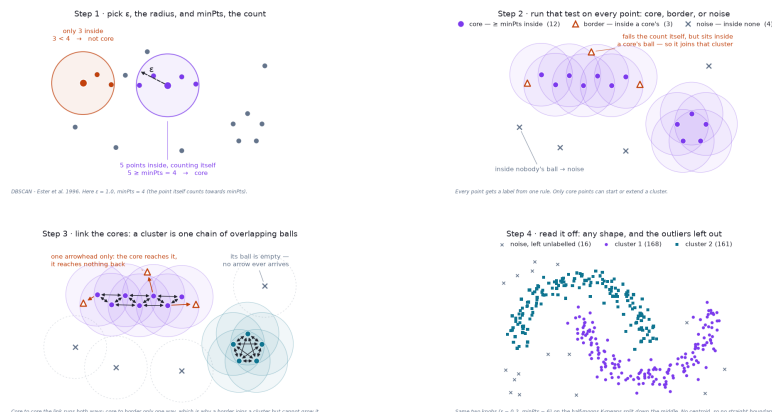


Figure 11 Core to core the disk runs both ways; core to border only one way, which is why a border joins a cluster but cannot grow it.

The algorithm in four panels: (a) the two knobs, drawn as circles of radius ϵ with a count of neighbours (one point passes the minPts test, the other does not; (b) every point classified as core (circles), border (triangles) or noise (crosses); (c) core points within ϵ of each other are chained (double-headed arrows), border points hang off the chain (single-headed); (d) the result on two interleaved half-moons with uniform noise) both crescents recovered whole, the scattered points discarded.

breadth-first search (every core point inside the current frontier's ϵ -neighbourhood joins the cluster and becomes part of the frontier, and border points are attached as they are met. The growth stops when the frontier is exhausted. The result does not depend on the order in which points are visited, except for the assignment of border points that are within ϵ of two different clusters) an ambiguity the original paper leaves to the implementation, and a real reproducibility hazard when comparing libraries.

Cost: with a spatial index, $O(n \log n)$ on average; without one, $O(n^2)$. The clustering itself is single-pass and cheap; the work is in the neighbourhood queries, which is the k NN machinery of §5 again.

6.3. Choosing the two knobs

minPts is the easier of the two. It sets the order of the smallest structure you are willing to call a cluster, and a common rule of thumb is $\text{minPts} \approx d + 1$ for d -dimensional data (here 17) or $\ln n$ for large samples. Our pipeline uses $\text{minPts} = 5$, which is deliberately permissive: it keeps small clusters such as the Pleiades candidates in play, at the cost of admitting noise-driven groups.

ϵ is the harder one, and the standard data-driven recipe is a k -distance plot: sort every point's distance to its k -th nearest neighbour, plot the sorted values, and look for the knee. The point where the curve leaves the bulk of the data and starts to climb. The knee is where the density threshold should sit. It is a heuristic, and it is exactly the step that fails in this project:

THE SINGLE-THRESHOLD PROBLEM

DBSCAN has one density threshold for the whole dataset. Our data have two

populations at very different densities: 357 056 field stars forming a broad, dense, chemically smooth background, and 25 clusters of 6–230 members each sitting inside it. Set ε for the clusters and the field becomes one enormous cluster containing everything; set it for the field and the clusters fragment into noise. There is no middle value that works, and the failure is not a tuning problem. It is the assumption of **Table 3** being false. The next two chapters exist because of it.

The scale of the effect is visible in the results: on our DR19 run, an un-normalised abundance space lets HDBSCAN merge the entire field into a single group (recall ≈ 1.0 with precision ≈ 0.001), which is the signature of a threshold set too high for the background. Row normalisation (§2) breaks that blob and is the single most effective precision lever in the pipeline.

EXERCISES

1. Draw the ε -ball classification by hand for the 2-D dataset in **Figure 11** with $\varepsilon = 0.35$ and `minPts` = 4, and mark every point core, border or noise. Then change `minPts` to 6 and say precisely which points change status.
2. Produce the k -distance plot for the DR19 member-plus-field matrix at $k = 5$ and locate the knee. Repeat for $k = 2$ and $k = 20$. How does the knee move, and what does that say about whether a single ε exists for these data?
3. Border points within ε of two core points of different clusters are assigned by visit order in the reference implementation. Find two such points in a small synthetic example, and construct two visit orders that produce different partitions. Why does this matter when comparing scikit-learn with a GPU implementation?

7. HDBSCAN*: A HIERARCHY OVER EVERY DENSITY

Deck: slides 22–26: mutual reachability, the tree, and the long-lived branches.

If the problem with DBSCAN is one global density threshold, the obvious repair is to use all of them at once and let the data choose. That is what HDBSCAN* does (Campello, Moulavi & Sander 2013; McInnes, Healy & Astels 2017): it builds a single hierarchy of density levels, scores each candidate cluster by how long it survives as the threshold moves, and returns the candidates that persist. It is the clusterer attached to t-SNE and UMAP in our pipeline, and (in a different guise) it is inside EVOC.

7.1. From points to a density-aware metric

The construction starts by making the metric itself density-aware. For each point, compute the core distance $\kappa_k(\mathbf{x})$ of §5: the distance to its k -th nearest neighbour. Then replace the original distance between two points with the *mutual reachability distance*,

$$d_{\text{mreach}}(\mathbf{a}, \mathbf{b}) = \max \{ \kappa_k(\mathbf{a}), \kappa_k(\mathbf{b}), d(\mathbf{a}, \mathbf{b}) \}. \quad (4)$$

Read it as a floor: two points in a dense region are pushed apart to at least the local density scale, so a sparse point can never be “close” to anything. Under this metric, a

dense region becomes a compact object even when its members are far apart in the original geometry, which is exactly the property DBSCAN lacked.

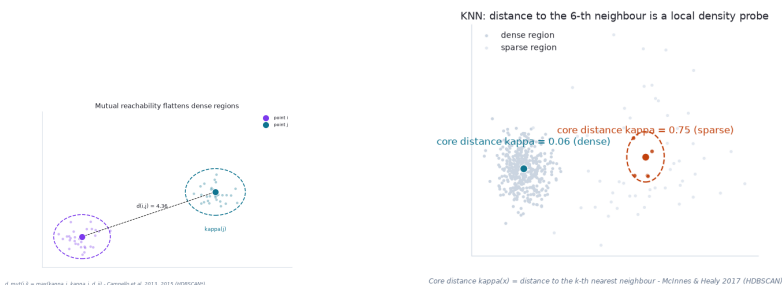


Figure 12

The ingredients of **Equation 4**, drawn in the deck. Left: two points, each ringed by its own core distance, with the straight-line distance between them; the mutual reachability distance is the largest of the three quantities. Right: the core distance itself (the radius that contains exactly k neighbours) which is the local density estimate of §5 written as a length.

7.2. Building the hierarchy

With the metric fixed, the rest is single-linkage clustering at every threshold simultaneously, computed efficiently:

1. Build the minimum spanning tree (MST) of the mutual-reachability graph. One tree contains every single-linkage merge at every density level: the largest edge weight on the path between two points is the density at which they merge.
2. Sort the tree's edges by weight and cut them in that order, producing a dendrogram of nested partitions: the density hierarchy (**Figure 13**).
3. Condense: a cluster that splits into pieces, one of which is smaller than `min_cluster_size` and short-lived, is treated as noise rather than as a new cluster. This is what makes the tree robust against single-linkage's notorious chaining.
4. Score every remaining branch by *stability*: the sum, over its members, of the difference between the density at which the member leaves the cluster and the density at which the cluster was born (**Equation 5**). Selecting the set of branches with the largest total stability (choosing a parent over its children when the parent scores higher) gives the final labels.

$$S(C_j) = \sum_{\mathbf{x} \in C_j} (\lambda_{\mathbf{x}} - \lambda_{\text{birth}}(C_j)), \quad \lambda = 1/d_{\text{mreach}}. \quad (5)$$

7.3. What the knobs mean

HDBSCAN* replaces DBSCAN's two knobs with two different ones, and both are easier to reason about:

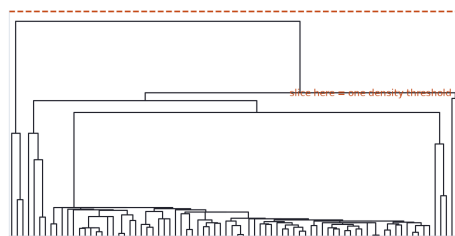
- `min_cluster_size` is a persistence floor, not a density threshold. It says: a structure smaller than this many points is not evidence of a cluster, so collapse it into its parent rather than report it. Our default is 5 (deliberately low, because several of our real

Mutual reachability:

The distance used by HDBSCAN*: the maximum of the two core distances and the true distance (**Equation 4**). It makes dense regions self-similar across scales.

Condensed tree:

The hierarchy after small clusters have been progressively collapsed into their parents as the density threshold rises. Its branches are the candidate clusters, weighted by lifetime (persistence).



HDBSCAN* - Campello et al. 2013, 2015. Single linkage on mutual-reachability distances.

Figure 13

The density hierarchy as a single-linkage dendrogram with a horizontal density cut. Everything above the line is one cluster; everything below it has split. HDBSCAN* does not commit to the line: it scores every branch by how far it survives above its own birth level and keeps the branches with the best scores.

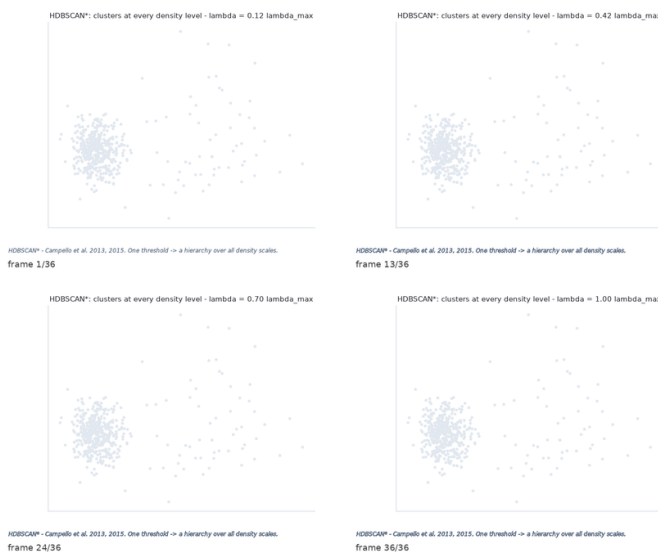


Figure 14

The same data at four density levels, from the deck's animation: clusters appear as the threshold drops, some grow, and some merge into each other. HDBSCAN* sees the whole sequence at once and keeps the objects that live long enough to be structure rather than coincidence.



Figure 15

One density level of the sweep. Density-based clustering is not a fixed partition of the data but a family of partitions indexed by scale; the algorithm's job is to pick from the family without being told the scale in advance.

clusters are small) and §13 reports the effect of raising it to 10, which on the DR17 data was the single largest improvement in recall.

- `min_samples` sets the k inside the core distance of **Equation 4**: how many neighbours define the local density scale. Larger values give a smoother, more conservative density estimate. The usual bias–variance trade of §5.
- The metric is a third, silent knob: the whole construction presumes that a distance is meaningful. Cosine and Euclidean give different hierarchies; that is why §2 normalises rows, and why the fairness argument for comparing methods rests on the metric being shared.

7.4. Why it fits our problem, and where it still fails

The hierarchy is a natural fit for a field of 25 clusters of very different sizes: it can, in principle, keep a 6-member cluster at one density and a 230-member cluster at another. In practice two things still bite.

First, `min_cluster_size` is a *single* size floor. It cannot be right for the Pleiades with 23 candidates and for M 67 with 230 members at the same time: one of them will be over-merged or the other fragmented. The next chapter removes this knob entirely.

Second, and more fundamentally, the algorithm inherits the geometry it is given. On our real abundances the residual failure is not the hierarchy but the space: the field is a single dense blob that contains the clusters, and no choice of `min_cluster_size` separates an object from its own background. That is what motivates the embeddings of §10–§12: change the geometry, then let HDBSCAN* work as designed.

EXERCISES

1. For the four points **a**, **b**, **c**, **d** of a square grid, compute d_{mreach} with $k = 2$ and verify that it is a valid metric. Which of the metric axioms does it break if you omit the κ terms and use $\max\{d(\mathbf{a}, \mathbf{b}), \text{const}\}$?
2. Compute the stability score of **Equation 5** for a two-branch tree in which one branch contains 100 points that survive two decades of λ and the other contains 10 points that survive five. Which does the algorithm prefer, and does that match your intuition about which is a better cluster?
3. Run HDBSCAN* on the DR19 member-plus-field matrix with row normalisation on and off. Record the number of clusters and the largest cluster fraction in each case, and identify which setting produces the “one blob” failure of §6.

8. PLSCAN: DROPPING THE SIZE FLOOR

Deck: slides 27–31: peaks, persistence, and the bars that survive.

HDBSCAN* removed DBSCAN’s density threshold but kept a single global `min_cluster_size`: a point count, below which a structure is treated as noise. That floor is a bad fit for our sample, where clusters range from 6 to 230 members. PLSCAN (persistent multiscale density-based clustering, Bot, McInnes & Aerts (2025)) takes the last step and removes it, replacing “is this group big enough?” with “how long does this group persist?”.

Persistence: The range of minimum cluster sizes over which a cluster exists as a separate structure, $s_{\max} - s_{\min}$ (Bot, McInnes & Aerts 2025). A long-lived peak is structure; a short-lived one is noise. It is measured in stars (the same units as the size floor it replaces) which is why a 6-member cluster can outlive a 60-member fluctuation.

Barcode: One horizontal bar per cluster, its length equal to its persistence, drawn on a common scale axis. Reading the barcode is the whole selection step.

8.1. Scale-space, and what persistence measures

The idea comes from topological data analysis. Rather than committing to one density threshold, consider the whole family of density estimates obtained by smoothing the data at every scale. A real cluster appears as a *peak* that stays separate from its neighbours as the scale changes; a fluctuation appears as a peak that is absorbed almost immediately. The range of scales over which a peak exists as an independent object is its *persistence*, and it is a statement about the data’s structure rather than about a count.

The algorithm builds this structure explicitly:

1. **Scale-space.** Estimate the density field over a range of scales (equivalently: over a range of effective minimum cluster sizes), tracking the peaks at each scale.
2. **Leaf tree.** Record the peaks and their merges as a tree, where a *leaf* is a density peak that exists in its own right: the analogue of the branches of the HDBSCAN* condensed tree, but indexed by scale rather than by k -th-neighbour distance.
3. **Persistence trace.** For each leaf, measure the interval of scale over which the cluster survives as a separate object. Clusters by construction cannot be shorter-lived than their own persistence, which is what makes the criterion self-consistent.
4. **Selection.** Rate every cut of the leaf tree by the total persistence of the leaves it keeps, and take the cut with the highest total persistence; read off the labels at that cut.

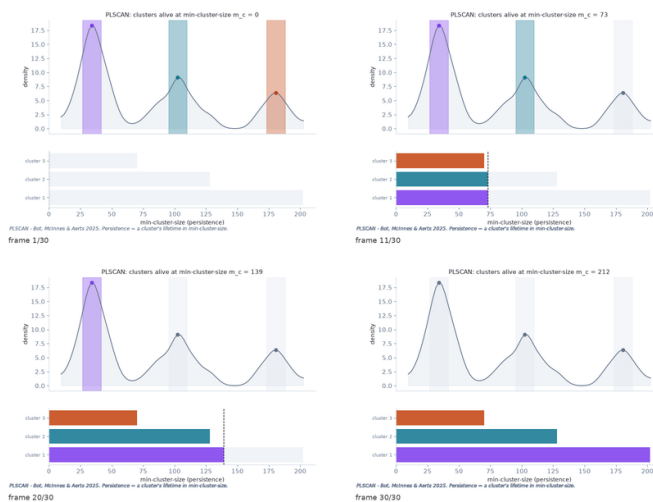


Figure 16

The persistence sweep, four frames of the deck’s animation: the minimum-cluster-size axis is swept, and each peak’s bar grows for as long as that peak remains a distinct cluster. Bars that stop early are fluctuations; bars that run the length of the sweep are structure.

8.2. What changes, and what does not

Two honest statements, because the summary of this method is often oversold.

What changes. The criterion for “is this a cluster?” is no longer a single size. A 6-member cluster that is a strong, long-lived density peak survives; a 60-member fluctuation

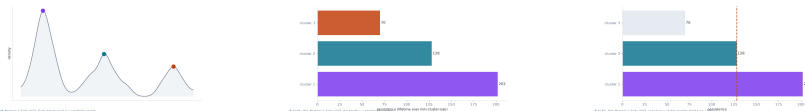


Figure 17

The three moves: (a) find the density peaks; (b) measure each peak's persistence and draw the barcode; (c) keep the bars of the cut with the largest total persistence. The barcode replaces the size floor with a quantity measured over a range of sizes; the neighbour count k that defines the core distance is still a choice, and when the trace has several peaks, so is which peak to report.

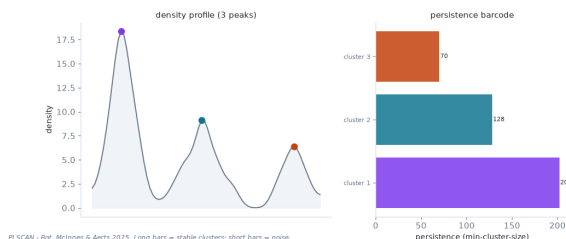


Figure 18

A three-peak density profile and its persistence barcode. The vertical extent of each bar is the range of “minimum cluster size” over which the corresponding cluster exists; the horizontal position identifies the peak. A min-cluster-size floor would cut all three bars at the same height, which is exactly the assumption PLSCAN avoids.

that dissolves as soon as the scale changes does not, because the cut through the leaf tree is chosen by the trace over all cuts rather than fixed in advance. That is the behaviour our sample needs, where M 67 (230 members) and NGC 2158 (6 members) must be judged by the same rule. What changes is not the units but who chooses: persistence is measured in stars, and the data pick the value.

What does not change. The cut is still a choice. PLSCAN makes it automatically, by maximum total persistence, not without assumptions. The neighbour count k that sets the core distance is still an input, and their own sensitivity study is the evidence that this matters less than HDBSCAN*'s size threshold. The assumption is moved to a quantity whose meaning is more stable across datasets, and it is made explicit in the barcode, where a user can see what a different cut would cost. Nor does any of this repair a bad space: if the field and the clusters are not separated in the geometry, no persistence criterion will separate them.

8.3. Where it appears in this workbook

PLSCAN enters our pipeline through EVoC, and that is worth understanding before §12. EVoC's final step sweeps the clustering of its internal node embedding across scales, scores each resulting layer by the total persistence of the clusters it contains, and returns the best-scoring layer's labels. That is the PLSCAN idea used as a *model-selection rule* inside a larger algorithm. The reason EVoC needs no `min_cluster_size` from the user even though the density clustering inside it would otherwise want one. On the deck's own terminology: the first two columns of the EVoC pipeline come from UMAP's graph machinery, the middle from HDBSCAN*, and the last from PLSCAN.

EXERCISES

1. Sketch a one-dimensional density with three peaks of different heights and widths. For each peak, mark the interval of smoothing scale over which it survives as a separate peak, and draw the resulting barcode. Now add a fourth peak that is real but tiny: 2% of the mass of the smallest other peak. Does the barcode distinguish it from a fluctuation?
2. In HDBSCAN* terms, what is the difference between “this branch is shorter than `min_cluster_size`” and “this branch is shorter-lived than the others”? Construct a dataset where the two rules give different answers, and say which you would trust.
3. The persistence of a cluster is a range of minimum cluster sizes, i.e. a number of stars (their Eq. 7). Using the member counts of **Table 1**, work out what a persistence of a given value implies for M 67 and for NGC 2158, and argue whether the same cut can serve both.

9. VALIDATION: KNOWING WHEN A SCORE LIES

Deck: slides 19–20 (internal and external validation) and slide 3.

In a supervised problem, validation is arithmetic: hold out labels, count mistakes. In clustering there is nothing to hold out, and the score you report is chosen by you. This chapter is therefore the most important one in the workbook, and the one in which we report the most unflattering numbers, including two of our own that forced us to withdraw a published claim.

Homogeneity: How pure each predicted group is: 1 when every cluster an algorithm reports contains a single true cluster, even if that true cluster is split across several predictions.

Completeness: How intact each true cluster is: 1 when every true cluster ends up in one predicted group, even if that group contains several true clusters. Homogeneity and completeness trade off.

9.1. Internal indices: is the partition tidy?

Three classical measures, all computed from the data and the partition alone.

- **Sum of squared errors.** The K-means objective of **Equation 1** (§4). Monotone in the number of groups, so it is only useful as a curve against K , never as a score.
- **Silhouette** (Rousseeuw 1987). For a point i , let $a(i)$ be its mean distance to the other members of its own group and $b(i)$ its mean distance to the members of the nearest other group. Then $s(i) = (b(i) - a(i)) / \max\{a(i), b(i)\} \in [-1, 1]$, and the silhouette is the mean over points. It rewards compact, well-separated groups, which is a statement about geometry, not about truth.
- **Dip test** (Hartigan & Hartigan 1985). A significance test for unimodality of a one-dimensional distribution. Its role here is upstream of clustering: it answers whether a variable (or a projection) shows any multi-peaked structure at all. If nothing is multimodal, the clusters an algorithm returns are imposed by the algorithm.

All three share one weakness, and it is fatal if forgotten: *a tidy partition of unstructured data is easy to produce*. The silhouette of a K -means solution on uniform noise is not zero, and the dip test is one significance test among many you could run. Internal indices are for diagnosing the behaviour of an algorithm, not for certifying a discovery.

9.2. External indices: comparison with something the algorithm never saw

Here the labels come from outside the algorithm. Two families are used in this workbook, and we use them for different tasks.

Cluster-only separation follows Garcia-Dias et al. (2019). Take only the known members, cluster them, and compare the predicted grouping with the true one using an information-theoretic pair plus an assignment score:

$$h = 1 - \frac{H(C | K)}{H(C)}, \quad c = 1 - \frac{H(K | C)}{H(K)}, \quad v = \frac{2hc}{h + c}, \quad (6)$$

where C are the true cluster labels, K the predicted ones, and H the Shannon entropy; accuracy is the fraction of correctly assigned stars after matching predicted to true labels with the Hungarian algorithm (maximising the diagonal of the confusion matrix). The pairing is not cosmetic. A method that merges every cluster into one group has $c = 1$ and $h \approx 0$; a method that shatters them has $h = 1$ and low c . Quoting one without the other is how a broken result reads as a good one, and we do it ourselves in **Table 4** for exactly that reason.

Field retrieval follows the operational question: for each cluster, take the predicted group that overlaps its true members most, and measure

$$\text{recall} = \frac{\text{true members recovered}}{\text{all true members}}, \quad \text{precision} = \frac{\text{true members in the group}}{\text{stars in the group}}. \quad (7)$$

Recovery fraction (Casamiquela et al. 2021) answers the question a per-cluster study actually asks: how many of the known clusters came back? For each true cluster, take the predicted group that holds most of its stars (the best-overlap rule above). The cluster counts as *recovered* at a threshold θ when that group holds at least θ of the cluster *and* at least θ of the group is the cluster. Their per-cluster completeness and homogeneity, our recall and precision, both at once:

$$\text{RF}_\theta = \frac{\#\{\text{clusters recovered at } \theta\}}{\#\{\text{clusters}\}}, \quad \theta = 0.40 \text{ or } 0.70. \quad (8)$$

Its companion is the *statistical fraction*: the share of predicted groups that recover no cluster at all, which is how the other paper measures the amount of decoration in its own solution. We adopt this pair for the same reason we adopt their task: of all the scores in this chapter it is the one whose chance level is near zero whatever the sizes and number of clusters, so it survives comparison between different samples. The implementation is `cluster.baseline.recovery_fraction`, and it reproduces their Table 2 exactly, which is the only external check on it available.

Two complements to **Equation 7** appear in the results. *kNN purity* (§5) is parameter-free cohesion, immune to the best-overlap protocol below. *AUROC* answers a refusal question rather than a discovery one: given a true member and a contaminant, how often does the method rank the member as more plausible? It is used in §13 to show that the rejection power of spectra grows with cluster age.

THE “BEST OVERLAP” IS A BEST CASE

The protocol above scores a method by its single best-matching predicted group per cluster. That is deliberately generous (it is the only fair way to compare algorithms that return different numbers of groups) but it means a method can look good by returning many groups, only one of which overlaps each cluster. It also cannot be gamed into positive precision: the denominator is the size of the predicted group, so a group with 3 true members in it and 300 field stars scores 0.01 however well it “recovered” those three.

9.3. Five ways a score lies

Each of these is a case we hit. They are ordered from the oldest to the most recent, and the fourth is the reason the validation chapter is in the middle of the workbook rather than at the end.

1 · The degenerate partition.. A clustering that returns two groups with two thirds of the stars in one of them produces a homogeneity of 0.269, a floor set by the collapse, not a measurement of the method. Two of our PCA rows sat on that floor, and a first draft of this work described the spectral latent as “three times better than PCA”. It was three times better than a crash. The check that catches it is a column nobody prints by default: the fraction of stars in the largest predicted group. We now print it, and **cluster head-to-head** flags degenerate rows automatically.

2 · Comparing different populations.. The abundance arm of our first head-to-head was scored on 25 clusters and 1 215 stars, while every other arm was scored on 5 clusters and 55 stars. The abundance row therefore looked worse than the others for a reason that had nothing to do with abundances: it was being asked a harder question. The fix is structural, not statistical. Intersect the sets of stars before scoring, so every row of the table describes the same 982 stars and the same 25 clusters. Two tables in this workbook have been rebuilt that way, and the rebuilt numbers are the ones quoted.

3 · Row order, and duplicate stars.. Two checks on the same member matrix, both of which should have been run before any of it was published.

Row order. Permuting the rows of the abundance matrix (a transformation that leaves every pairwise distance exactly unchanged) moves the t-SNE homogeneity between 0.218 and 0.560, and the number of predicted groups between 2 and 22. The sorted order gives 0.560; the mean over eight random orders is 0.268 ± 0.100 . The geometry is identical in all nine runs; what changes is the optimiser’s path, which depends on initialisation and on the order in which the Barnes–Hut approximation accumulates interactions. Every other arm is stable to a few hundredths: UMAP on the same abundances 0.579 ± 0.009 , and the masked-AE latent 0.740 ± 0.003 under t-SNE. The lesson is not “t-SNE is broken” but “report the row-order spread of any embedding with a stochastic initialisation”, because on small samples it can exceed the effect you are claiming.

Duplicates. An audit of the same matrix found that it contains repeat entries for the same star: of 1 002 member rows, 807 carry distinct APOGEE_IDS and 34 carry none, so 332

rows share an identifier with another row (`scripts/population_counts.py`). The copies are not identical (the median spread between two rows of the same star is 0.04 dex per element, comparable to the abundance differences we are trying to measure) and they are concentrated: M 3 contributes 102 duplicate rows, M 67 ninety. This inflates the member counts of **Table 1** (they are row counts, not star counts), double-weights those clusters in every macro average, and is the most likely explanation for the fragility of the abundances-only t-SNE row above. The clean fix is to collapse each star’s rows before clustering.

4 · Tuning on the score.. The lever sweep of §13 chose element weights by maximising the very metric it then reported, on a single cluster at a single seed. That is a legitimate way to *find* a lever and an illegitimate way to *quantify* it, and the current run exposed the difference: the same switch that lifted t-SNE recall on the DR17 data ($0.09 \rightarrow 0.44$) lowers precision on the DR19 data ($0.83 \rightarrow 0.24$). A lever is a hypothesis; only a re-run on the current data, quoted with both recall and precision, turns it into a result.

5 · The chance level of a mutual-information score moves with the sample.. Small groups inflate h , c and V . Shuffle Casamiquela et al. (2021)’s 175 stars at random into groups with the sizes they observe (31 clusters, most of them tiny) and the V-measure is 0.51; do the same on our 982 stars in 25 clusters and it is 0.06–0.12. Their published $V = 0.55$ therefore sits 0.04 above its null, and our $V = 0.58$ sits about 0.47 above ours. The two numbers are not comparable, and a table that prints them side by side without their nulls asserts a comparison that does not exist. **Equation 8** does not have this problem: chance level 0.04 for their sample shape and ≤ 0.001 for ours. This is the reason §13.6 is written in recovery fractions.

9.4. The protocol this workbook follows

Out of all that, six rules, which the numbers in §13–§15 obey:

1. **Same population or no comparison.** Arms are intersected on the stars they share before any score is computed.
2. **Mean $\pm$ standard deviation over seeds.** Seven seeds (42, 0, 1, 2, 7, 13, 99), never a single run, and never a best-of.
3. **Kinematics are the ceiling, not a competitor.** They define the labels; the interesting question is always how close chemistry or spectra get to that ceiling.
4. **Parameter-free scores alongside tuned ones.** kNN purity and AUROC cannot be improved by tuning, which makes them the sanity check on everything else.
5. **Report the degenerate rows.** Largest-group fraction and number of predicted groups are printed next to every score, so a collapse is visible without recomputation.
6. **Recompute the chance level before comparing with a published score.** A score quoted without its null is a number without units, and the null moves when the sample does (rule 5 above).

EXERCISES

1. Show that a clustering that puts every star in a single group has $h = 0$ and $c = 1$, and that a clustering that gives every star its own group has $h = 1$ and $c \approx 0$.

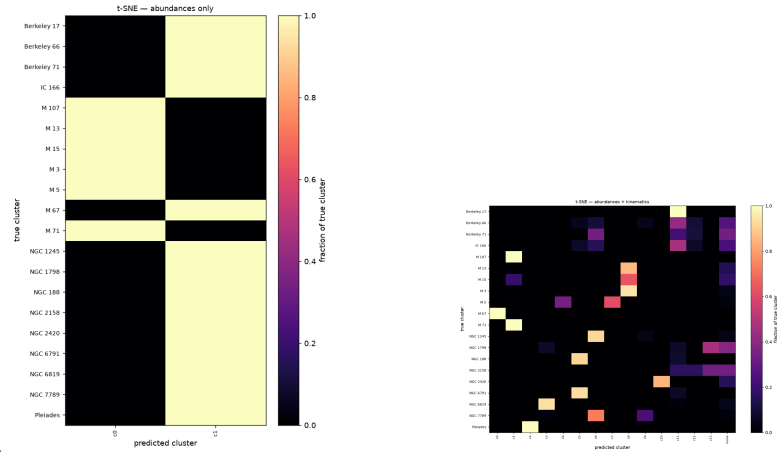


Figure 19

The confusion matrix of a t-SNE clustering of the members, with the true cluster on one axis and the predicted group on the other (**Equation 6** visualised). Left: abundances only. One predicted group absorbs most of the true clusters, which is the completeness-rich, homogeneity-poor blob described above. Right: adding kinematics. The blocks along the diagonal are what “the clusters are separated” looks like in a matrix, and the difference between the two panels is the central result of §13.

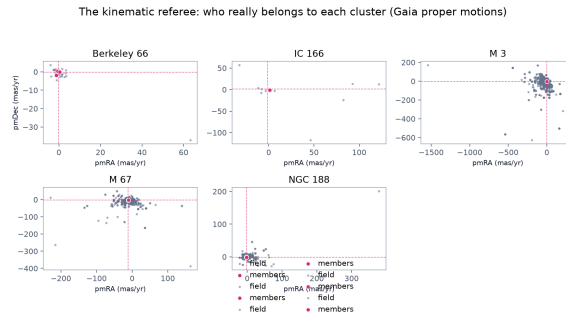


Figure 20

Why kinematics can referee. Each panel shows proper motion in right ascension and declination for the stars around one cluster; the tight group sharing a velocity is the cluster, the surrounding cloud is the field. These are the observables that define the labels, so they can never be an input to a method that is scored against them.

Where does accuracy sit in each limit?

2. Reproduce the row-order check: cluster the DR19 member matrix with t-SNE + HDBSCAN nine times, once sorted and eight times with a random row order, fixing the seed. Report the spread of homogeneity and of the number of predicted groups. Now do the same with the masked-AE latent. Why is one stable and the other not?
3. Collapse the duplicate rows of the member matrix (for example by taking the median abundance vector per `APOGEE_ID`) and re-run the cluster-only table. Which rows of **Table 4** move, and does M 3’s share of the sample change the conclusion of the

ablation?

- Design a σ -clipped chemical membership and use it to score a kinematic clustering. Based on §2, explain in three sentences why the resulting number would be meaningless, and what the equivalent mistake would look like in your own field.

10. t-SNE: A MAP OF NEIGHBOURHOODS

Deck: slides 32–38: the two probability models and the descent between them.

Everything so far has treated the 16-dimensional abundance vector as a space in which distance means something. That assumption is weaker than it looks. In high dimensions, distances concentrate: the ratio of the farthest to the nearest neighbour of a point tends to 1, so “close” and “far” describe almost the same thing and a density-based method loses contrast. t-SNE (van der Maaten & Hinton 2008), building on stochastic neighbour embedding (Hinton & Roweis 2002), answers this by giving up on preserving distances at all and preserving *neighbourhoods* instead. Then drawing the result in two dimensions.

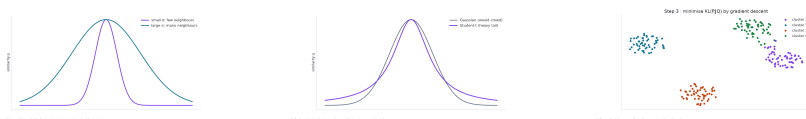


Figure 21

The three moves of t-SNE, from the deck: **(a)** in the high-D space, convert distances to neighbour probabilities (here the point’s neighbourhoods are drawn as spheres of decreasing probability); **(b)** in the 2-D map, do the same conversion with a heavy-tailed kernel; **(c)** move the points downhill on the Kullback–Leibler divergence until the two sets of probabilities agree as closely as they can.

10.1. From distances to probabilities

For each point i , t-SNE defines the probability that j is its neighbour in the high-dimensional space as a Gaussian in the distance,

$$p_{j|i} = \frac{\exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|\mathbf{x}_i - \mathbf{x}_k\|^2 / 2\sigma_i^2)}, \quad (9)$$

and symmetrises, $p_{ij} = (p_{j|i} + p_{i|j}) / 2n$. The novelty is σ_i : each point gets its *own* bandwidth, chosen so that the neighbourhood distribution has a fixed entropy. The user-facing parameter is that entropy, expressed as *perplexity*,

$$\text{Perp}(P_i) = 2^{H(P_i)}, \quad H(P_i) = -\sum_j p_{j|i} \log_2 p_{j|i}, \quad (10)$$

found by binary search on σ_i . Perplexity is the effective number of neighbours each point is required to have, and it is the parameter you must justify. van der Maaten & Hinton (2008) suggest 5–50; a widely used default is $\sqrt{n}$ or $n/100$; our pipeline’s default is 30, and the swap to 50 measurably improved the layout for our $\sim 5\,000$ -star samples.

Perplexity: The effective number of neighbours each point is given in t-SNE’s high-dimensional probability distribution (**Equation 10**). It is the only required hyperparameter, and it is a statement about the scale of structure you are looking for.

Crowding problem: In a low-dimensional map there is not enough room to place moderately distant points at moderate distances. t-SNE’s heavy-tailed low-dimensional kernel fixes it by making the map’s tail fat.

10.2. The low-dimensional half, and the crowding problem

The map is a set of 2-D positions $\mathbf{y}_i$. Their neighbour probabilities use a Student- t kernel with one degree of freedom,

$$q_{ij} = \frac{(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|\mathbf{y}_k - \mathbf{y}_l\|^2)^{-1}}, \quad (11)$$

and the layout is found by minimising the Kullback–Leibler divergence $D_{\text{KL}}(P \parallel Q)$ by gradient descent. The heavy tail in **Equation 11** is the whole point. With a Gaussian map kernel, the volume available at radius r in two dimensions grows only as r , while the volume of a shell in 16 dimensions grows as r^{15} : a point that should be moderately distant from its neighbours has nowhere to go, and the optimiser either piles dissimilar points on top of each other or flings them arbitrarily far away. This is the crowding problem, and the t_1 kernel solves it by allowing large distances to be cheap: $q_{ij} \sim 1/\|\mathbf{y}_i - \mathbf{y}_j\|^2$, so a point can sit far from another at almost no cost in divergence. The resulting gradient has an intuitive form (each point is pulled towards its high-D neighbours and pushed away from the others, with the repulsion weak at long range) and it is what **Figure 22** animates.



Figure 22

The descent, four frames of the deck’s animation: from the initial (PCA) layout, through the early exaggeration phase in which clusters coalesce while empty space opens up between them, to the settled map. The exaggeration phase is a device, not physics: it deliberately over-states attraction at the start to escape a poor initial arrangement.

HOW TO READ A t-SNE MAP

The picture is a map of neighbour probabilities, and the following are all *not* readable from it. (i) Distances between clusters: only the relative order of neighbourhoods is preserved, so a gap can appear between two groups whose members are mutual

neighbours. (ii) Cluster sizes and densities: t-SNE normalises local density away, so a tight blob and a loose one can have the same number of members. (iii) Global geometry: rotation, reflection and the arrangement of clusters are arbitrary. (iv) The absence of a gap: a method that must place every point somewhere will place well-separated points on the same island if the perplexity is wrong.

10.3. What t-SNE does not do

It does not cluster. The output is a 2-D embedding, and the visual groupings are an artefact of a continuous optimisation. The workflow that has become standard (and that our pipeline uses) is to treat the embedding as a feature space and cluster *that* with a density-based method, normally HDBSCAN* (§7) or EVoC (§12). The distinction matters when quoting results: a t-SNE row in our tables is really “t-SNE followed by HDBSCAN*”, and both steps contribute to the score.

Three further practical facts, each of which we have hit:

- **Initialisation changes the answer.** The original implementation started from a random layout; the modern default starts from a PCA of the input. Both are stochastic in practice, and on our abundance matrix the difference is not subtle: see the row-order audit of §9, where the same data gave homogeneity between 0.218 and 0.560 depending on the order of the input rows.
- **The distance metric is fixed to Euclidean.** t-SNE has no metric parameter, which is why §2’s decision to make Euclidean distance and cosine similarity equivalent on our matrix is not a detail: it is the only reason the t-SNE, UMAP and EVoC arms are comparable at all.
- **Cost grows faster than linearly.** The original $O(n^2)$ formulation is unusable past a few thousand stars; the Barnes–Hut approximation (van der Maaten 2014) brings it to $O(n \log n)$ per iteration, at the cost of the order dependence noted above. Our field runs embed 30 107 stars in this regime.

10.4. t-SNE on real clusters

The published anchor for this section is Kos et al. (2017), who applied t-SNE to GALAH abundances (13 elements, calibrated with The Cannon) and recovered 7 of the 9 clusters in their sample, the Pleiades among them, as coherent groups in a two-dimensional map. **Figure 23** reproduces their figure, and it is worth being precise about what it shows: a known cluster, its members in red, sitting inside a map built from the surrounding field. It is a positive result, and it is also a reminder of the asymmetry in this literature: the clusters were selected because they were already known, and the method is judged on whether it finds them again.

Our own cluster-only results for the abundance matrix illustrate the failure mode the caveats above predict, and they do it in two registers. The single run of **Table 5** gives the t-SNE arm the highest completeness of the three methods (0.879) and the lowest homogeneity (0.292), the signature of a map in which most clusters have been drawn into one region. The seven-seed average of **Table 4** looks better (0.560 homogeneity, 0.601 completeness), and §9 shows that the same matrix moves between 0.218 and 0.560 on homogeneity, and between

2 and 22 groups, when only the order of the input rows changes. The neighbourhoods are preserved, and the neighbourhoods were not separated to begin with.

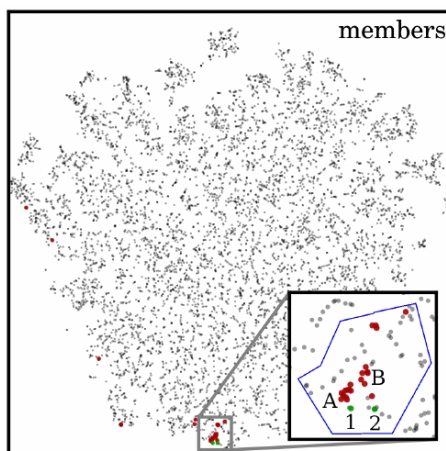


Figure 23

t-SNE of GALAH abundances around the Pleiades, as published by Kos et al. (2017) and used in the lecture: the red points are stars known to be cluster members, and the inset zooms in on where they concentrate, marking the two groups the method finds (A, B) and the two member candidates it adds (1, 2). A known cluster, recovered from field data. The benchmark that strong chemical tagging has to beat.

EXERCISES

1. Implement the binary search for σ_i in **Equation 10** that fixes perplexity at a target value for every point. Plot the resulting σ_i against local density on the DR19 matrix. Does the adaptive bandwidth do what it is supposed to do?
2. Embed the member matrix with perplexities 5, 30 and 100, and for each compute the kNN purity of §5 plus the cluster-only homogeneity. Which perplexity would you choose, on which evidence, and does the choice change the paper's conclusion?
3. Take the t-SNE embedding of the abundances and run HDBSCAN* with `min_cluster_size = 5, 15 and 50`. Report the number of groups and the largest group fraction each time. Identify which run is degenerate and say what the degeneracy looks like in the map.
4. The low-dimensional kernel is $q_{ij} \propto (1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}$. Replace it with a Student- t kernel of $\nu = 5$ degrees of freedom, re-derive the gradient, and explain what you expect to change in the map's long-range behaviour.

11. UMAP: A GRAPH, THEN A LAYOUT

Deck: slides 39–42: graph, fuzzy edges, layout.

t-SNE converts a fully connected distance matrix into probabilities and then optimises a two-dimensional layout against them. UMAP (McInnes, Healy & Melville 2018) takes a different route to a similar-looking picture: build a weighted nearest-neighbour graph,

decide what that graph’s low-dimensional analogue should look like, and find positions that make the two agree. The construction is more explicit about its assumptions, it is faster, and (as §13 shows) it behaves differently on our data in a way that teaches something about both methods.

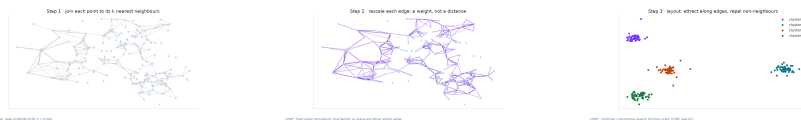


Figure 24

UMAP in three moves, from the deck: **(a)** build the k -nearest-neighbour graph of the high-dimensional data; **(b)** weight its edges by local distance and treat the result as a fuzzy set. Every edge has a probability of existing; **(c)** lay the graph out in low dimensions so that strong edges are short and the layout stays spread out, by minimising the cross-entropy between the two edge distributions.

11.1. The high-dimensional graph

For each point i , let ρ_i be the distance to its nearest neighbour (a local connectivity term that makes the metric robust in dense regions) and let σ_i be a bandwidth chosen so that the total edge weight out of i is a fixed constant:

$$w_{ij} = \exp\left(-\frac{d(i, j) - \rho_i}{\sigma_i}\right). \quad (12)$$

Small distances between close neighbours are saturated at $\exp(0) = 1$; the bandwidth is solved per point by binary search, exactly as t-SNE solves for σ_i in **Equation 9**, with the target here being $\sum_j w_{ij} = \log_2 k$ rather than a perplexity. The result is a directed graph, made symmetric by the fuzzy-set union $w_{ij} + w_{ji} - w_{ij}w_{ji}$.

Two design decisions are worth naming. First, the graph is built from an approximate k NN search (NN-descent, §5), which is why UMAP scales to 10^6 points where t-SNE does not. Second, the graph is *local*: edges exist only to the k nearest neighbours, so UMAP never sees a long-range distance. Every claim about UMAP preserving “more global structure” than t-SNE has to be read against that fact. The long ranges it appears to preserve are produced by the layout’s initialisation and by the repulsion term, not by information in the graph.

11.2. The low-dimensional half

In the map, the probability assigned to an edge is a parameterised curve

$$q_{ij} = \left(1 + a \|\mathbf{y}_i - \mathbf{y}_j\|^{2b}\right)^{-1}, \quad (13)$$

with a and b fitted so that the curve approximates the shape of the high-dimensional weights for the requested `min_dist`. The objective is the cross-entropy between the two edge distributions, minimised by stochastic gradient descent with negative sampling: a handful of random non-edges attracted or repelled per real edge, which is the second reason UMAP is fast.

n_neighbors: The size of the local graph, and UMAP’s analogue of perplexity: it sets the scale of structure the layout can see. Small values concentrate on very local structure; large values give a more global, blobier picture.

min_dist: How tightly points may pack in the layout. It changes the appearance of density, not the structure found: a low value makes clusters look like tight clumps.

11.3. Parameters, and why our defaults are what they are

UMAP exposes more knobs than t-SNE, and three of them matter here.

- **n_neighbors** (default 15 in our pipeline, 15 in the library). On the DR17 lever sweep this was the difference between a working and a broken layout: at **n_neighbors** = 5 the recall on M 67 fell from 0.415 to 0.071, while at 30 it was indistinguishable from the default (0.410). The lesson generalises: too few neighbours gives a graph that fragments the field, not one that isolates the cluster.
- **metric**. Our matrix is L2-normalised (§2), so Euclidean and cosine distances are monotonically related and the metric should be nearly irrelevant. In the same sweep, forcing **metric=cosine** cut recall from 0.415 to 0.044. The nominal geometry did not change; the implementation's internal handling of the graph did. Report the setting you used.
- **random_state**. UMAP's optimisation is genuinely stochastic: unlike the t-SNE rows in our tables, which came out identical across the seven seeds we tried, UMAP's homogeneity spread is ± 0.011 on the masked-AE latent. That is small, but it is the same order as some of the gaps quoted between arms.

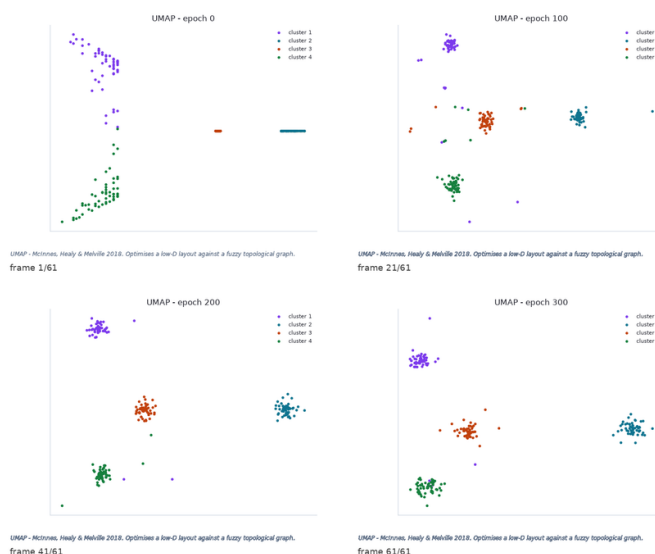


Figure 25

The layout being optimised, four frames of the deck's animation: the graph is embedded with attraction along the edges and repulsion between sampled non-neighbours, and the structure emerges as the cross-entropy falls. There is no phase analogous to t-SNE's early exaggeration; the layout is driven the whole way by the same objective.

11.4. How it differs from t-SNE in practice

The two methods answer the same question (produce a low-dimensional map that keeps neighbours together) and disagree in ways that matter for the rest of this workbook:

- **Speed and scaling**. UMAP is the only arm we can run on the full 357 056-star field

without sampling; the density-clustering arm of our pipeline therefore defaults to it for field work, and the t-SNE arm is run in region mode or on subsamples.

- **What “noise” looks like.** UMAP has no noise model. Everything gets a position, so a cluster in a UMAP map is a density peak and nothing more, which is why the pipeline clusters the embedding afterwards rather than reading groups off the picture.
- **Blob formation.** On our abundance matrix, UMAP reaches a higher cluster-only homogeneity than t-SNE (0.578 vs 0.560) while producing larger, fewer groups, and in the field-retrieval task it recovers all 978 members at a precision of 0.0013, that is, by declaring a large part of the field to be cluster members. **Table 7** is the clearest illustration in the workbook of why recall and precision must be read as a pair, and of why the “which method is better” question has no answer until the task is specified.

EXERCISES

1. Implement the per-point binary search for σ_i in **Equation 12** that fixes $\sum_j w_{ij} = \log_2 k$, and verify numerically on a random dataset. Where does the local-connectivity term ρ_i matter most: dense regions or sparse ones?
2. For a small dataset, compute UMAP’s edge weights and then run the layout twice: once with PCA initialisation and once with a random one. Do the two layouts agree on which points are neighbours? Which of them would you publish, and what would you have to state about it?
3. `min_dist` changes the appearance of the map without changing the graph. Using a fixed embedding, vary `min_dist` over $\{0.0, 0.1, 0.5, 1.0\}$ and compute the cluster-only homogeneity each time. Does the clustering result move, and does that support or undercut the use of UMAP maps as evidence?
4. UMAP’s repulsion term acts on sampled non-edges, so its notion of “far” is an artefact of negative sampling. Construct a small dataset in which the layout places two true clusters far apart and the reverse, by changing only `random_state`. What does that imply for reading distances from a published UMAP figure?

12. EVoC: ONE FIT, THREE STAGES

Deck: slides 43–45: the fusion, the pipeline, and why C-space.

The three methods so far are composed by hand: t-SNE or UMAP produces an embedding, then HDBSCAN* (or a persistence-driven variant of it) produces labels, then the user decides how the two are coupled. EVoC (Tutte Institute for Mathematics and Computing 2024) removes the seam. It takes the graph, embeds it, clusters it, and chooses the clustering itself, in a single fit, and it is the method our pipeline uses for its production runs.

Read the deck’s pipeline figure first: the point of the method is that the three stages share one objective rather than being tuned against each other.

12.1. Stage 1: the graph, built on the metric you meant

EVoC’s first stage is a k -nearest-neighbour graph (§5) constructed directly from the input representation, with a configurable metric. This matters in our setting for a specific reason:

persistence, from the same graph that the embedding was built on. The distinction is not about quality but about where the arbitrary decisions live, and it is why our benchmark reports all three, since each one answers a different question about the same data.

12.3. Stage 3: layers, persistence, and no `min_cluster_size`

The clustering stage produces not one answer but a family of them, indexed by scale. Each candidate layer is scored by the total persistence of the clusters it contains (the PLSCAN criterion of §8) and the best-scoring layer’s labels are returned. Consequences that matter in practice:

- **No size floor to set.** The user is never asked for `min_cluster_size`, which is the single knob that most changed results in our DR17 sweeps (§13: raising it from 5 to 10 moved t-SNE recall from 0.087 to 0.404). EVoC replaces it with an internal persistence criterion, which is more stable across datasets but less inspectable: you cannot see which layers were considered.
- **The number of clusters is an output.** On the field runs, EVoC returns a few dozen groups on average, none of them large: one of the reasons its recall is high and its precision catastrophic (Table 7).
- **The returned layer is a model-selection decision.** Persistence is a principled criterion, not a guarantee: if the field and the clusters are not separated in the graph’s geometry, the best-scoring layer is the one that splits the field most persistently.

12.4. What EVoC does for us, and what it does not

Positive first. EVoC is the only arm that needs no post-hoc clustering choice, it works directly on the cosine geometry that our data actually have, it is the cheapest of the three to run on large samples, and on cluster-only separation it is competitive: homogeneity 0.420 ± 0.043 on abundances, 0.733 ± 0.008 on the 64-dimensional PCA latent, both measured over seven seeds on the same 982 stars as the other arms.

Now the part that took us a while to accept. Two results in the literature around this method, and two of ours, all point the same way:

WHAT WE HAVE TO SAY AGAINST OUR OWN FAVOURITE

The seed spread is larger than several of the gaps. EVoC’s homogeneity on abundances is 0.420 ± 0.043 over seven seeds: a spread that overlaps with the t-SNE and UMAP arms. In the five-cluster head-to-head the spread is ± 0.08 – 0.11 . Any ranking that depends on differences of that size is not a ranking.

The self-supervision claim is a t-SNE/UMAP result, not an EVoC result. On the corrected same-population table, the masked autoencoder scores 0.70 ± 0.08 against 0.64 ± 0.08 for abundances on EVoC (overlapping within errors) while the *supervised* CNN wins EVoC outright at 0.80 ± 0.03 on the four-cluster sample. We withdrew the sentence that said the abundance-free latent wins everywhere.

High recall at near-zero precision. On the all-sky field-retrieval task, EVoC recovers 0.557 of the true members at a precision of 0.0033: the groups it returns are mostly field stars, by two orders of magnitude. It is the same failure that UMAP shows in that table, and it is the reason the pipeline’s field work runs a rejection stage after

the clustering rather than trusting the labels.

EXERCISES

1. EVoC clusters an embedding built from the graph, and HDBSCAN* clusters the embedding built by t-SNE or UMAP. Explain, in three sentences each, why the first is more self-consistent and why the second is easier to diagnose when it fails.
2. On the member matrix, run EVoC with the seed varied over the same seven values used in this workbook, and report homogeneity as mean $\pm$ standard deviation. Now do the same for t-SNE with a fixed seed but nine row orders (§9). Which source of variability is larger, and what does that say about which number belongs in a paper?
3. Take the EVoC labels on the all-sky field run and compute, for each returned group, the fraction of its members that are known field stars. Plot that distribution. How many groups would you have to ignore before the remaining ones have precision above 0.5?
4. The persistence criterion selects a clustering layer without user input. Construct a two-dimensional dataset with a dense uniform background and one small dense cluster, and report which layer EVoC selects. Does the criterion find the cluster, the background, or something else?

13. THE BENCHMARK: WHAT WE MEASURED

Deck: slides 46–47 for the benchmark and slide 48 for the lessons in short form.

This chapter is the case study: the methods of §4–§12 run on the data of §2, scored by the protocol of §9. The order of business is the benchmark grid, then the two tasks separately, then the levers, and finally the list of things this benchmark cannot tell you.

13.1. The design

Three things vary, and nothing else:

- **Signal.** The 16 standardised abundances of §2; a 64- or 256-dimensional PCA projection of the APOGEE spectra (§14); a 256-dimensional masked-autoencoder latent of the same spectra; the four kinematic observables (parallax, two proper motions, radial velocity); and concatenations of chemistry with kinematics.
- **Method.** t-SNE, UMAP and EVoC, each followed by HDBSCAN* clustering where the method needs one, with the defaults of **Table 2**.
- **Task.** *Cluster-only separation:* cluster the known members among themselves and ask whether the clusters come out distinct, scored with **Equation 6**. *Field retrieval:* search the full sample for members and score with **Equation 7**.

Three conventions keep the comparison honest, and all three are choices a reader should check before believing any row:

1. **Same population.** Every arm in **Table 4** is scored on the same 982 stars and the

same 25 clusters. An earlier version of this table compared arms scored on 25 clusters against arms scored on 5, which is the mistake of §9.

2. **Seven seeds, reported as mean $\pm$ s.d.** The kinematics-only t-SNE row is deterministic, several other rows are not, and the difference is information rather than noise to be hidden.
3. **Two referees.** Field retrieval is scored against two independent definitions of “member”: the kinematic selection of §2, and the Simbad cluster memberships of the literature. Every conclusion below holds under both; the Simbad referee moves precision by -0.01 to -0.08 (the largest shifts are in the 64-dimensional PCA arms) and changes no ranking.

13.2. Task 1: separating the clusters from each other

Table 4 is the central result of the workbook. Read it by column first: the columns are methods, the rows are the signal the method was given.

signal	dim	t-SNE	UMAP	EVoc
abundances	16	0.560 ± 0.000	0.578 ± 0.011	0.420 ± 0.043
PCA of spectra	64	0.741 ± 0.000	0.754 ± 0.007	0.733 ± 0.008
PCA of spectra	256	0.688 ± 0.003	0.685 ± 0.014	0.653 ± 0.020
masked AE latent	256	0.740 ± 0.002	0.760 ± 0.011	0.714 ± 0.036
kinematics only	4	0.942 ± 0.000	0.940 ± 0.005	0.876 ± 0.012
abundances, K-means (K known)	16	0.449 (single run)		

Table 4 Cluster-only homogeneity on the same 982 stars in 25 clusters, mean $\pm$ standard deviation over seven seeds. Higher is better. The K-means row is the classical baseline of §4, run on the 1002-row member matrix with $K = 25$ and $n_{\text{init}} = 10$ by `article/scripts/kmeans_baseline.py`: it is not an embedding, so it has no column to sit in, and it is quoted for scale rather than for ranking. The recovery fractions of the same arms, and the comparison with the published result, are in Table 10.

Four things in that table are worth stating out loud.

Chemistry is not nothing. All three methods reach homogeneity well above the trivial floor of $1/25 = 0.04$: the 16 abundances carry real cluster information, and the embedding methods extract some of it. This is the optimistic half of the chemical-tagging programme, reproduced on data that post-date the paper that reported it.

Chemistry is not enough. The same methods on four kinematic observables (a quarter of the dimensions) reach 0.94. Nothing about the abundance arms is competitive with kinematics, and no amount of embedding repairs it: the four-dimensional signal is simply more informative about which cluster a star belongs to.

Dimension helps, up to a point. Going from 16 abundances to the 64-dimensional PCA latent raises EVoc’s homogeneity from 0.42 to 0.73 and t-SNE’s from 0.56 to 0.74. Going on to 256 principal components *lower* it (0.69 for t-SNE, 0.65 for EVoc) which is the bias–variance trade of §3 in its least forgiving form: the extra 192 directions carry variance, not signal, and every one of them is a direction in which the field can become an apparent neighbour.

One row of that table is not a measurement. The abundances-only t-SNE row is unstable under row order, as the audit of §9 showed: the same matrix gives 0.218–0.560

depending on how the rows are ordered, and the single-run baseline of **Table 5** gives 0.292 for what is nominally the same experiment. Any claim that rests on the difference between 0.29 and 0.56 for t-SNE on raw abundances is built on the input ordering.

features	t-SNE	UMAP	EVOC
abundances (16-d)	0.292	0.547	0.409
abundances + kinematics	0.767	0.748	0.680

Table 5 The single-run baseline of the DR19 re-run (cluster baseline, no seed averaging, the full 1 002-row member matrix). It is included because it is the number our own documentation quotes, and because comparing it with Table 4 is instructive: concatenating kinematics lifts every method by 0.25–0.48, and the abundances-only t-SNE row differs between the two tables by more than the seed spread of any other row.

signal (excluding M 3)	t-SNE	UMAP	EVOC
abundances (884 stars)	0.199 (degenerate)	0.581	0.499
masked AE latent (884 stars)	0.711	0.742	0.658

Table 6 Ablation on the 982-star population: the globular cluster M 3 contributes 98 of the stars (884 remain) (or 154 of the 1 002 member rows of Table 1) and it is the largest globular in the sample, though not the largest cluster: M 67 supplies 218 stars and the Pleiades 113. Remove M 3 and the abundances-only t-SNE arm does not merely change, it collapses: two groups with 75% of the stars in the largest, zero stars labelled noise. The spectral latent is unaffected by the same ablation. A result that depends on one cluster being present is not a measurement of the method. The per-cluster counts of both populations are in results/population_counts.csv (python scripts/population_counts.py -baseline-min-finite 9).

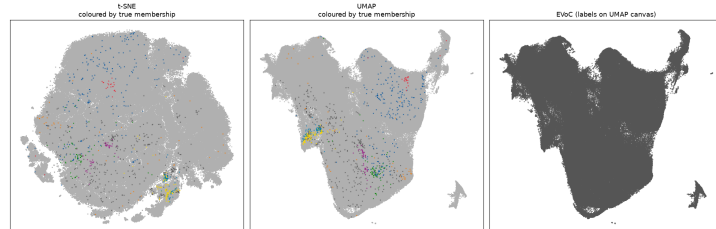


Figure 28

The whole grid in one figure: method (rows) against signal (columns), for the two tasks. Read the left block for cluster-only separation and the right block for field retrieval, and notice that the two blocks do not agree on a winner, no single row is best at both. This is the figure the lecture ends the benchmark on, and it is the reason §13 refuses to name a best method.

13.3. Task 2: finding members in the field

Now the operational question. Take 30 107 stars (the 25 clusters plus a stratified sample of the field), cluster them, and ask how many of the known members were recovered and how much of what was returned was real.

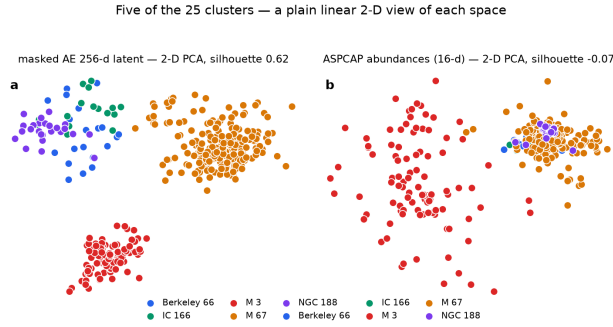


Figure 29

The head-to-head on 64-dimensional PCA spectra, cluster-only separation, seven seeds. The error bars are the seed spread: the masked-AE and PCA-64 arms overlap, which is the tie reported in §14 and the reason the spectral result is stated as “spectra beat abundances, and do not beat PCA”.

signal	method	recall	precision	kNN purity
abundances (16-d)	t-SNE	0.201	0.240	0.124
abundances (16-d)	UMAP	1.000	0.001	0.075
abundances (16-d)	EVOC	0.557	0.003	(
PCA 64-d	t-SNE	0.417	0.631	0.528
PCA 64-d	UMAP	0.374	0.592	0.434
PCA 64-d	EVOC	0.585	0.003	)
masked AE 256-d	t-SNE	0.418	0.619	0.504
masked AE 256-d	UMAP	0.385	0.513	0.419
masked AE 256-d	EVOC	0.255	0.005	:

Table 7 Field retrieval, all-sky, on the same 30 107 stars and the same 25 clusters, scored against the kinematic membership (978 members). Recall and precision are the best-overlap definition of Equation 7; kNN purity is the parameter-free cohesion score of §5, and it is undefined for EVOC because that arm returns no embedding to measure it in. Scoring against the Simbad referee instead shifts every precision by -0.01 to -0.08 and changes no ranking.

The three lessons here organise most of what the rest of this workbook says about chemical tagging in practice.

Recall without precision is a statement about the field, not about the clusters. UMAP on raw abundances recovers *every* true member: at a precision of 0.001. It has not found the clusters; it has labelled a large part of the field as cluster, and the best-overlap protocol dutifully reports that the members are inside the resulting group. EVOC does the same thing more politely (0.557 recall, 0.003 precision). This is the trap that the literature’s habit of quoting a single number walks into, and it is why **Table 7** prints three.

Spectra read directly beat element ratios, and it took a chemical ablation to be sure. The 64-dimensional PCA latent and the 256-dimensional masked-AE latent both reach recall ≈ 0.42 at precision ≈ 0.62 from the abundances’ 0.20/0.24. Two independent representations of the same spectra agree with each other to within 0.002 in recall and 0.012 in precision, which is the honest way to say that the improvement is real while the difference

between the 64- and 256-dimensional spectral rows is not.

The gap to the ceiling is the story. Kinematics alone recover 0.93 of the members at a precision limited by velocity doppelgängers rather than by the method (§3). Chemistry, read as well as we can read it today, gets to 0.42 recall. Strong chemical tagging (recovering a dispersed cluster with no kinematic help) is not what these numbers demonstrate; what they demonstrate is that modern self-supervised spectral representations find *something* real about a cluster in a spectrum, and that the something is diluted by a factor of two relative to parallaxes and proper motions.

13.4. The curse of dimensionality, measured

Concatenating a high-dimensional, weakly informative signal onto a low-dimensional, strongly informative one does not add their information. It dilutes the strong signal, because distance is shared across dimensions. The DR17 study measured this on 878 members in region mode with the kinematic labels held out of the ground truth:

signal (dims)	cluster homogeneity	field recall	field precision
abundances (16)	0.25 / 0.51 / 0.51	0.73	0.47
spectral latent (256)	0.40 / 0.69 / 0.64	0.72	0.51
+ kinematics (20 / 260)	0.62 / 0.74 / 0.72	0.83 / 0.80	0.63 / 0.61
kinematics only (4)	0.97 / 0.96 / 0.92	0.93	0.63–0.67

Table 8 Signal comparison on the DR17 data (three numbers per cell are the t-SNE / UMAP / EVoC homogeneity). Kinematics alone near-solve the cluster-only task and set the field-recall ceiling; adding chemistry on top improves the cluster-only score slightly and never improves recall beyond what kinematics already achieved. The DR19 re-run reproduces the pattern with different absolute values (kinematics 0.94, abundances 0.56, abundances + kinematics 0.77 over the same 982 stars).

The mechanism is worth stating precisely, because it is the reason this workbook spends three chapters on geometry rather than one on algorithms. A distance in a 20-dimensional space is an average over 20 coordinate differences. If 16 of those coordinates are chemistry and carry weak information about cluster identity, while 4 are kinematics and carry strong information, then the strong coordinates are outvoted, not mis-weighted, outvoted, because no reweighting recovers information that the metric has already blurred. Rescaling chemistry *up* to compensate costs precision in the field, as §13’s lever table shows. The correct response is architectural, not numerical: cluster on the strong signal, then apply the weak signal as a rejection test. That is the two-stage pipeline, and it is the single most successful idea in this project.

13.5. Levers, and a lesson about bug fixes

The DR17 study swept the pipeline’s knobs on one cluster (M 67) inside a 30° region, with 5 000 stars and 183 referee members. Every run logged its full configuration. The table is worth reading as a list of *hypotheses*, not results, and the section that follows explains why.

The lesson is not that hyperparameters matter, which is known, but that the *largest single effect in a sweep can be a bug*. A configuration flag that does nothing is worse than a missing flag, because it produces a negative result that looks like an experiment. The DR17 re-check of the same lever on DR19 data did not reproduce the recall gain, which

lever	recall	prec.	UMAP recall	verdict
baseline	0.087	0.117	0.415	:
<code>min_cluster_size</code> 5 → 10	0.404	0.129	0.525	best single
perplexity 30 → 50	0.383	0.124	0.415	free win
perplexity 30 → 10	0.033	0.353	0.415	precision for recall
<code>SNR_MIN</code> 100 → 150	0.296	0.089	0.310	recall up, precision down
dwarf-only sample	0.149	0.165	0.158	best purity
drop S, K, V, Mn	0.055	0.256	0.038	hurts recall
UMAP <code>metric=cosine</code>	0.087	0.117	0.044	hurts UMAP
UMAP <code>n_neighbors</code> = 5	0.087	0.117	0.071	worse
UMAP <code>n_neighbors</code> = 30	0.087	0.117	0.410	= baseline
element weights (after a bug fix)	0.437	0.158	0.339	best lever
weights + size floor	0.432	0.160	0.525	not additive

Table 9 DR17 lever sweep (M 67, 30° region, 5 000 stars, 183 referee members, seed 42). The recall and precision columns are t-SNE; the last two rows exist because of a bug: the element-weighting switch was originally a no-op, since the weights were applied and then undone by the standardisation step that followed. Reordering the two operations turned the largest apparent non-effect into the largest effect in the table. Two rules of thumb survive from this sweep (the size floor of 5 fragments the field, and a perplexity below $N/100$ starves the embedding) and one warning: the best levers do not compose.

is exactly what the protocol of §9 predicts for a lever that was selected on one cluster at one seed: selecting on a metric and reporting the same metric is not validation. The DR19 region sweep of the same switch confirms the warning from the other direction. A recall that climbs to 0.96 while precision falls to 0.05 is a cluster that has grown to swallow the field.

13.6. Where this sits in the literature: Casamiquela et al. (2021)

Every number so far in this chapter has been scored against our own baselines. The closest published measurement is Casamiquela et al. (2021), and it is close enough to sit in the same table: the same metric triple, the same task, and a sample built to be the best case that spectroscopy can produce. They measured 16 elements differentially in 175 red-clump members of 31 thin-disc open clusters, with an internal coherence of about 0.03 dex per element and *no field stars*. They then clustered the abundances directly with HDBSCAN (no embedding, no projection) and fine-tuned its free parameters to maximise the number of correctly recovered clusters (their Sect. 4.2). They call the sample itself their *best-case scenario*. Their published result is $h = 0.49$, $c = 0.63$, $V = 0.55$, $\text{RF}_{40} = 0.29$ (9 of 31 clusters) and $\text{RF}_{70} = 0.03$ (one cluster, NGC 2682); their own summary of that run is that more than 70 % of the groups it finds are statistical, mixing stars from several real clusters.

Four things follow from that table.

Their clustering step recovers less on our stars than on theirs. Run their configuration on our 982 stars and the recovery fraction falls to 0.08 (0.11 on the open clusters alone, both being Berkeley 17 and NGC 1245), against 0.29 on their own sample; under the standardised variant it returns 168 groups and recovers nothing at all. Two differences remain that we cannot separate: our abundances are survey-quality (~ 0.05 dex

sample	$n_{\star} / n_{\text{cl}}$	h	c	V	RF ₄₀	RF ₇₀
Casamiquela et al. (2021), published	175 / 31	0.49	0.63	0.55	0.290	0.030
their clustering step, our stars (raw dex)	982 / 25	0.499	0.416	0.454	0.080	0.040
their clustering step, our stars (standardised)	982 / 25	0.475	0.407	0.438	0.000	0.000
our pipeline: t-SNE	982 / 25	0.560	0.601	0.580	0.320	0.080
our pipeline: UMAP	982 / 25	0.578	0.572	0.575	0.326	0.006
our pipeline: EVoC	982 / 25	0.420	0.529	0.466	0.189	0.006
our pipeline: K-means (K known)	982 / 25	0.454	0.490	0.471	0.080	0.000
open clusters only, UMAP	665 / 18	0.573	0.422	0.486	0.405	0.040
open clusters only, t-SNE	665 / 18	0.471	0.458	0.465	0.222	0.000
open clusters only, EVoC	665 / 18	0.520	0.419	0.464	0.214	0.048

Table 10 The published comparison. Rows 2–3 are *their* clustering step (HDBSCAN on the abundances, min_cluster_size=2, their 4×7 parameter grid, best grid point by RF₄₀) run on our stars; their configuration is defined in dex, so the raw-abundance row is the faithful replication and the standardised row is a rescaling check. Their 4×7 grid never reaches RF₄₀ > 0.11 on our stars. Rows 4–10 are our arms, mean over seven seeds (the h , c , V columns repeat Table 4); the open-cluster subset is the closest analogue of their sample. Chance levels, from shuffling the true labels into groups of the observed sizes: for their sample shape $V = 0.51$ (0.44–0.51 across the three partition shapes tested) and RF₄₀ = 0.04 (95th percentile 0.10); for their clustering step run on *our* stars $V = 0.29$ (their rule cuts 165 groups, and many small groups inflate the null) with RF₄₀ ≤ 0.001; for our own arms $V = 0.06$ –0.13 and RF₄₀ ≤ 0.001. Source: results/casamiquela_comparison*.csv and results/casamiquela_protocol.csv; the second one also carries the grid search.

per element against their 0.03) and our clusters are not a single evolutionary stage.

Our two-stage pipeline lands in the same range as their one-stage pipeline. t-SNE and UMAP on 16 survey abundances, with default HDBSCAN, recover 0.32 and 0.33 of the 25 clusters, and on the 18 open clusters alone (the subset closest to their sample) UMAP reaches 0.41. The honest reading is not that we beat them. It is that embed-then-cluster on coarser abundances reaches the same recovery as cluster-the-abundances on the best abundances available, and that neither gets past about a third of the clusters.

The failures have the same shape. Their surviving clusters sit at the edges of the abundance distribution (NGC 6705, NGC 2682, and NGC 2420, which they call the most metal-poor cluster in their sample. NGC 2420 is a regular recovery of ours too: UMAP recovers it in every seed of the open-cluster subset and EVoC in six of seven. Our arms otherwise recover the same handful of clusters in every seed) Berkeley 17, IC 166, M 71, NGC 1245, NGC 6819, NGC 7789, the Pleiades, and that set contains M 3, which §2.5 introduces as the metal-poor outlier of the sample. Whether this is the same edge-of-distribution mechanism their paper describes is a hypothesis this workbook does not settle: member evolutionary state is a confound we have not yet removed, which is what the DWARF_ONLY lever of §16 is for.

Their field-star test is the published analogue of our Task 2, and it agrees. Adding field stars to their sample leaves one cluster above the 40% threshold (NGC 188) and none at 70%. Our field retrieval (Table 7) agrees in direction and goes further: the best abundance-only arm reaches recall 0.20 at precision 0.24, and the arms that reach

high recall do it by labelling most of the field as cluster. Their element tests agree with ours too: restricting the set to 8 or 7 elements did not improve their recovery, and our dropped-element row in **Table 9** loses recall as well.

13.7. What this benchmark cannot tell you

Five limits, in descending order of how much they should trouble a reader.

1. **The sample is small and uneven.** 982 stars across 25 clusters is a few tens of stars per object, and the distribution is wide: M 67 supplies 218 stars and the Pleiades 113, while five clusters supply fewer than ten each once the duplicate rows are collapsed (Berkeley 17, NGC 1798, King 5, NGC 2158, King 7 (**Table 1** counts rows, and lists three). **Table 6** shows what a single well-populated cluster can do to a result, and every macro average in this chapter is therefore an average over a handful of clusters that are not equally measured. The member frame is itself configuration-dependent: the 1002 rows counted in **Table 1** need a finite-abundance floor of nine out of sixteen elements, while the shipped default of eight keeps 1215 rows) so the saved baseline scores are not reproducible from a bare `cluster baseline` run (`python scripts/population_counts.py -baseline-min-finite 9`).
2. **The member matrix contains duplicated stars.** 298 rows share an APOGEE_ID with another row, the duplicates differ by ~ 0.04 dex per element, and they inflate the largest clusters most. Until the sample is collapsed to unique stars, every macro average in this chapter is a weighted average over something that is not quite a star.
3. **Some rows are not measurements.** The abundances-only t-SNE row is row-order unstable (§9); the ablation row is degenerate by construction. Both are printed rather than dropped, because the pattern of which arms are fragile is itself a result: the spectral-latent arms are stable, and the raw-abundance t-SNE arm is not.
4. **The external comparison is one pipeline, not a survey of the literature.** §13.6 runs Casamiquela et al. (2021)’s clustering step on our stars and their metric triple on our scores, which is as far as a like-for-like comparison goes without their star list. We do not re-run the graph-attention autoencoder of Spina et al. (2025), which works on $\sim 47\,000$ APOGEE stars and deliberately adds kinematics and age to the chemistry, so any claim of state-of-the-art in this space should be read as unaudited.
5. **The counter-result stands, and it is not the same measurement.** Hogg et al. (2016) recovered phase-space structure by clustering 15 abundances with K-means, and our K-means row in **Table 4** matches EVOC’s homogeneity. On the recovery fraction it is the weakest arm of the table (0.08 against t-SNE’s 0.32), which is the difference between a tidy partition and a recovered cluster: *K*-means is told how many groups to make, so its homogeneity is high by construction, while the density methods have to find the groups. Whatever this workbook demonstrates about the limits of abundance-space clustering, it is a statement about APOGEE’s ASPCAP precision, these 25 clusters, and these methods, not a statement that chemical tagging does not work.

EXERCISES

1. Reproduce one row of **Table 4**: pick a signal and a method, run it over the seven

Masked

autoencoder: A self-supervised model that hides part of its input, encodes the visible part, and is trained to reconstruct the hidden part. It learns the structure of the data without any labels. In 1-D spectra, *how* you mask decides what it can learn.

Self-supervised:

Trained with a loss defined by the data itself (reconstruction) rather than by external labels. The property we care about here is negative: a self-supervised latent never saw an element ratio or a cluster membership, so it cannot be circular.

seeds, and report mean $\pm$ standard deviation. Compare with the published row. If your numbers differ, is the cause the population, the seed set, or a default?

2. **Table 7** reports precision for the kinematic referee. Recompute it against the Simbad memberships and plot the difference per cluster. Which clusters are most referee-sensitive, and does that correlate with their age or their $[\text{Fe}/\text{H}]$?
3. The ablation of **Table 6** removes M 3. Repeat it for the other clusters with more than 100 rows (M 67, NGC 6819, NGC 2243) and for the duplicates-cleaned matrix. How much of the instability survives cleaning?
4. §13.6 runs their pipeline on our stars but not ours on theirs. Their sample reachable star list is not public in a form we can re-run, so state what you would need from the other authors, and what a re-run on their stars would change about the conclusion of **Table 10**.

14. SPECTRA INSTEAD OF ELEMENT RATIOS

Deck: slide 47, the masked-autoencoder block and its four steps.

Everything in §13 treats a star as a 16-dimensional vector of abundances. That vector is a lossy summary of the observation it came from: a single APOGEE spectrum has 8575 wavelength pixels, and the abundance pipeline that turns those pixels into 16 numbers assumes a one-dimensional model atmosphere, a set of atomic and molecular line data, and a fitting procedure that was tuned on the Sun. This chapter asks the obvious question (if the spectra are the data, why cluster the summary?) and reports what happens when we do.

14.1. The baseline: PCA of the spectra

The honest first experiment is principal component analysis. Garcia-Dias et al. (2019) applied eight clustering algorithms to the 13 APOGEE abundances of each star under four dimensionality-reduction strategies, and this chapter asks the same question one level down: which representation of the observation keeps the cluster information. Reading spectra with learned models has a longer history in this field (the Cannon builds a data-driven spectral model from labels (Ness et al. 2015), and convolutional networks do the same job without the labels' linearity assumption (Ting et al. 2019)) and the question this chapter asks is which of those representations keeps the cluster information. Keep 64 components and the classification results of §13 already improve on the abundances; keep 256 and they get slightly worse. PCA is the right baseline and the wrong tool for one reason that is worth internalising: it maximises *variance*, and in a stellar spectrum the variance is dominated by things that have nothing to do with chemistry. The temperature-sensitive continuum shape, gravity-sensitive line profiles, interstellar reddening, and the signal-to-noise of the observation. A principal component that encodes T_{eff} is useful for almost every other problem in stellar spectroscopy and actively unhelpful here, because a hot field star and a hot cluster member will share it.

14.2. The model: a masked spectral autoencoder

Our proposal (He et al. 2022) is to train a self-supervised latent that has to predict the spectrum’s hidden parts:

1. Take the continuum-normalised spectrum, 8 575 pixels.
2. Mask 50% of the wavelength axis, in *contiguous blocks* of 200 pixels: roughly a spectral line region at APOGEE’s resolution.
3. Encode the visible pixels with five convolutional blocks, each halving the wavelength axis, with channel widths from 1 024 down to 64, ending in a global average pool and a 256-dimensional linear bottleneck.
4. Reconstruct the masked pixels with a mirrored decoder, and compute the loss *only where the input was hidden*.
5. Discard the decoder. Keep the encoder’s 256-dimensional output and feed it to the clustering pipeline exactly as the abundances were.

The model has 8.6 million parameters and trains in under an hour on one GPU on 39 945 field spectra plus 994 member spectra, with early stopping. Nothing in the loss function mentions abundances, clusters, kinematics, or age.

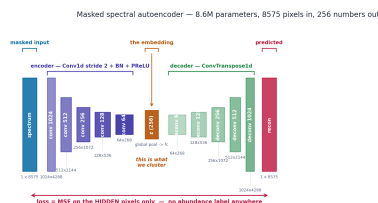


Figure 30

The architecture: five strided convolutional blocks compress the 8 575-pixel spectrum by a factor of 32 while widening the channel count, a global average pool removes the wavelength axis, and a linear layer produces the 256-dimensional latent. The decoder is a mirror of the encoder and is discarded at inference.

14.3. Why the masking is contiguous

This is the one design decision in the chapter that a reader should be able to justify from first principles, and it is where a naive implementation fails.

Mask single pixels at random and a convolutional network does not have to learn chemistry: the masked pixel’s immediate neighbours are visible, and the spectrum is smooth on the scale of a few pixels, so copying from the left and right solves the task. The reconstruction loss falls, the latent looks respectable, and the model has learned to interpolate. Mask *contiguous 200-pixel blocks* and the visible pixels adjacent to the hole are far enough away, in wavelength terms, to be in different lines; the model must use the global shape of the spectrum (which line regions are present, and how deep) to fill in the gap. That is where element ratios live.

14.4. What it achieves

On cluster-only separation it is the strongest signal we have (**Table 4**): homogeneity 0.740 ± 0.002 (t-SNE), 0.760 ± 0.011 (UMAP), 0.714 ± 0.036 (EVoC) on the 982-star

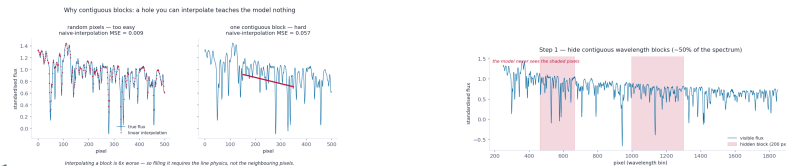


Figure 31

Left: why the masking is contiguous: a single-pixel hole is filled by interpolation from its immediate neighbours, while a 200-pixel block removes a whole line region and forces the model to use context. Right: the actual mask used (top) and the residual the model must predict (bottom), with the masked region shaded.



Figure 32

Encoding and reconstruction. Left: only the visible 50% of the spectrum enters the encoder. Right: the decoder's reconstruction of the hidden half, against the truth, with the loss computed only on the hidden pixels. The model is never rewarded for the parts it can see, which is what makes its latent a description of the spectrum rather than a copy of it.

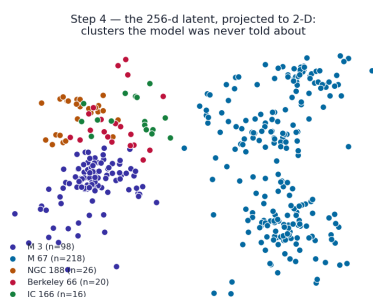


Figure 33

The 256-dimensional latent, projected. The spectral latent carries the abundance structure of the members because it had to reconstruct the lines those abundances are measured from, not because anyone told it she was looking at C, N, O or Fe.

population, against abundances 0.560/0.578/0.420. In the single-run baseline it reaches 0.789/0.874/0.770 on the full 1002-row matrix, and concatenating it with the abundances gives 0.879. The highest cluster-only score measured anywhere in this project. It also has the most important non-empirical property: because it never saw an abundance label, it cannot be circular in the way a supervised latent is.

On field retrieval it is a different story, and the difference is the most useful result in this chapter:

- **Facing the field, the masked latent does not beat PCA-64.** In Table 7 the two are 0.418/0.619 and 0.417/0.631: a tie to within the third decimal. Two very different representations of the same spectra (one unsupervised and linear, one self-supervised and deep) land on the same operating point.
- **The tie is the finding.** If a 64-dimensional linear projection reaches the same field-retrieval precision as a 8.6-million-parameter convolutional latent, then what

is limiting the precision is not the representation capacity of the model. It is the information in the spectra themselves, or the way our metric uses it.

- **EVoC disagrees, and we report both.** On the same cluster-only task with EVoC as the clusterer, the supervised CNN reaches 0.80 ± 0.03 on the four-cluster sample against the masked latent’s 0.70 ± 0.08 : the supervised model wins, and the two error bars overlap. We described the self-supervised result as a t-SNE/UMAP result, not as a universal one, and withdrew the stronger sentence.

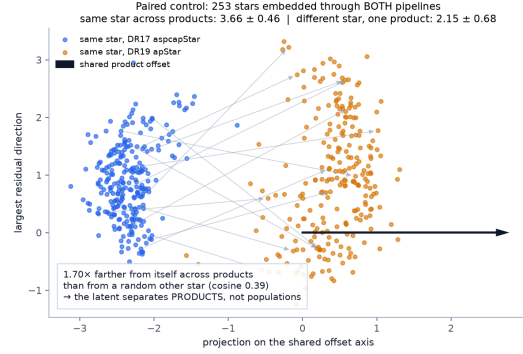


Figure 34

The paired control that ended the “two times purer” claim, which is the reason this figure is in the workbook. Each star is embedded through both pipelines; the panel shows the distance from a star to *itself* across data products against the distance to a different random star within one product. The mean is indicated. When a star is further from its own second embedding than from a stranger, the latent is encoding something about the data product rather than about the star.

14.5. The error we made, in full

A masked autoencoder trained on a mixed set of spectra will learn whatever distinguishes them, and our first training set was mixed in a way we did not notice. The field sample was 100% raw **apStar** spectra, while 91% of the member sample was continuum-normalised **aspcapStar** output, with median flux levels differing by a factor of ~ 6000 . Along that boundary the labels themselves are the difference.

The consequences, all measured:

- A linear probe recovers the data product from the latent at **AUC 0.999** on members alone, using 223 of the 256 dimensions ($p < 10^{-6}$).
- A one-bit flag (“which product is this?”) separates members from field stars at **AUC 0.957**: essentially the whole latent score, with no chemistry involved.
- The paired control of **Figure 34** is decisive: for 253 stars embedded through both pipelines, a star sits $1.70\times$ further from its own other-product embedding than from a random different star within one product (cosine distance 0.389, with 75% of the offset in one shared direction. The signature of a single global nuisance axis).

We first diagnosed this as a data-release effect (DR17 members, DR19 field). That was wrong, and the check that disproved it is worth repeating: DR19 reanalyses and includes DR17 (717 689 rows carry `release=‘dr17’`) and DR19 serves a uniform **mwmStar** spectrum

for **738 of 738** members in the backfilled list. The mismatch was between two products on the same sky, at the same epoch, from the same spectrograph.

So the field-retrieval row for the mixed latent is not interpretable, and the “twice as pure” precision claim that depended on it is withdrawn. What survives is narrower and solid: the cluster-only homogeneity, which is computed between members and therefore within one product, holds on a single-product subsample (0.612/0.686/0.575 against the abundances’ 0.484/0.550/ 0.488 on the same stars). The fix is not a caveat but a re-run: download uniform `mwmStar` spectra for every member and field star, re-embed, and repeat the benchmark. That run is staged in the repository (736 `mwmStar` spectra are already downloaded and waiting) and its results belong in the next version of this workbook rather than in this one.

Isochrone: A theoretical curve in the colour–magnitude diagram giving the luminosity and colour of every mass of a single-age, single-metallicity population. Fitting it to a cluster’s stars returns age, distance and reddening.

Distance modulus:
 $m - M = 5 \log_{10} d - 5$,
the difference between apparent and absolute magnitude. It is the fitted distance in a CMD; a 0.1 mag error is a 5% error in distance.

EXERCISES

1. Train the same autoencoder twice, once with single-pixel random masking and once with contiguous 200-pixel blocks, and compare the reconstruction loss on a held-out set. Then compare the two latents on the cluster-only task. Which model wins on reconstruction, and which on chemistry? Explain the disagreement.
2. Reproduce the linear probe of §14.5: fit a logistic regression from a 256-dimensional latent to a binary flag identifying the data product, and report the AUC on held-out member stars. Now repeat it for a latent trained on a single product. What is the AUC you would consider acceptable before publishing a field-retrieval number?
3. The paired control uses 253 stars with embeddings through both pipelines. Design an equivalent control for a study that only has one product, and state what it can and cannot detect.
4. The masked latent and PCA-64 tie on field retrieval. Working from the two latents, characterise what each one encodes: fit a linear model from each latent to T_{eff} , $\log g$ and $[\text{Fe}/\text{H}]$ in turn, and compare the cross-validated errors. Which representation spends more of its capacity on chemistry?

15. WHAT MEMBERSHIP IS FOR

Deck: slide 47, the isochrone and two-survey blocks.

Every number so far has been a score against labels. This chapter closes the loop: membership lists are not the product, they are the input to a measurement of what the cluster *is* (its age, its distance, its chemistry) and those measurements are what chemical tagging has to deliver if it is to matter for Galactic archaeology (Freeman & Bland-Hawthorn 2002; Bland-Hawthorn & Freeman 2016).

15.1. The measurement problem

Fitting an isochrone grid to a cluster means searching a three-dimensional parameter space (age, metallicity, distance modulus) for the combination that best reproduces the observed colour–magnitude diagram, with the reddening as a fourth, partly degenerate axis. We do it with the PARSEC grids through `ASteCA` (Perren, Vázquez & Piatti 2015), sampling the posterior with `emcee` (Foreman-Mackey et al. 2013). Two properties of that fit dominate



Figure 35

The two ends of the sample, which is why the same pipeline behaves differently on different clusters. Left: NGC 188 (old (7.6 Gyr), well populated, chemically distinct from the field. Right: IC 166) sparse and contaminated, with several member rows whose `APOGEE_ID` is blank (§9). In the first, the isochrone fit has hundreds of stars on a clean sequence; in the second, every contaminant moves the fit.

everything that follows.

Age, metallicity and reddening are partially degenerate. In the regions of the CMD our clusters occupy, an older, more metal-rich population at larger reddening can mimic a younger, metal-poor one at smaller reddening. The degeneracy is a property of the isochrone geometry, not of the data or the sampler: the posterior stays broad in the degenerate direction even when every star in the cluster is a perfect member. In the two-survey fit below, the age posterior width is $\sigma_{\log a} \approx 0.97$ (nearly a factor of ten in age) for a cluster whose distance is pinned to 0.01 mag.

Contaminants bias the parameters in a specific direction. Field stars in a cluster’s colour-magnitude diagram occupy the same locus as the cluster’s own stars in some mass ranges and a different one in others, and the bias they introduce is not random. Adding field stars to a metal-poor globular cluster lowers the inferred age; adding them to a young cluster raises it. Membership quality is therefore a parameter-estimation problem, not a bookkeeping one.

15.2. Distances from red clump giants

The most robust single measurement available to us is the distance, because red clump giants have nearly constant absolute magnitude: the median K -band magnitude of the members that fall in the $J - K$ clump slice is a standard candle with a known calibration. For the clusters where our membership is deep enough to have red clump members, the distances agree with the literature:

15.3. The case that decided us: NGC 2243 from two surveys

NGC 2243 is the demonstration that chemical tagging is a two-survey problem. APOGEE, being a near-infrared survey of the Galactic disc, observes a cluster’s *giants*: bright, plentiful, and excellent for distances and chemistry. GALAH observes at visible wavelengths and reaches the *main sequence and turnoff*, which is where age sensitivity lives. Neither survey alone constrains both.

The sequence we ran is the one a workshop should reproduce:

cluster	dm (clump)	dm (literature)	residual
NGC 6819	11.92	11.90	+0.01
NGC 2243	13.08	13.25	−0.16
NGC 7789	11.68	11.27	+0.41

Table 11 Red-clump distance moduli from member giants, against literature values. Two of the three agree to better than 0.2 mag, which for a membership list of a few dozen giants assembled from a survey without any prior membership information is the strongest validation in this chapter: the members we find are located where the clusters are.

1. Take the APOGEE member giants and fit an isochrone. The fit returns an age of 1.59 Gyr against a literature value of 1.07 Gyr (wrong by half) with a broad distance posterior ($\sigma_{dm} = 0.111$).
2. Use the red-clump distance (13.08 mag, agreeing with the literature value of 13.25 to within 0.17) as a tight prior, and select GALAH stars kinematically around the resulting centroid. At 4.7 kpc the parallax error rivals the parallax itself, so a relaxed parallax cut is required: a detail that took a re-run to get right.
3. Fit the combined main-sequence-plus-giants sample: age 0.98 Gyr (residual 0.09), $\sigma_{dm} = 0.086$.

A factor of six improvement in the age residual, from the same cluster and the same survey data, by adding the stars that are sensitive to age and prior-ing the distance with a measurement that is not. The age posterior remains broad ($\sigma_{\log a} \approx 0.97$), which is the degeneracy of §15.1 making itself felt, but the mode is right, and the distance is a fifth as uncertain.

15.4. The finale: better membership, better parameters

The clearest scientific payoff of the whole project is a single cluster. Fit an isochrone to three different membership lists for M 67: the literature values are age 2.82 Gyr, $dm = 9.54$, $[\text{Fe}/\text{H}] = 0.03$:

membership	n	age (Gyr)	residual
stage 1 (kinematic candidates, contaminated)	259	1.89	0.93
kinematic truth	219	2.45	0.37
two-stage (spectral rejection)	237	2.65	0.17

Table 12 Isochrone ages for M 67 from three membership lists, against the literature age of 2.82 Gyr. The two-stage pipeline (kinematic candidates, then rejection of spectral outliers) recovers the age better than the raw kinematic membership does.

Read the middle row carefully, because it is the surprising one. The kinematic membership is the *truth* for every benchmark in this workbook, and it still gets the age wrong by 0.37 Gyr. The two-stage list, which throws away 22 stars that kinematics accepted, gets it wrong by 0.17. The contaminants that the spectral stage removed were not random: they were young field stars with the right position and velocity, dragging the fitted age down.

Two honest qualifications. This is one cluster, so it is an existence proof that membership quality propagates into parameters, not a calibration of the size of the effect. And the effect

is measured in the same direction as the known bias of contamination on a young cluster's age, which is reassuring, and also means the mechanism is understood rather than merely observed.

15.5. What this does and does not establish

The Galactic-archaeology promise of chemical tagging is large: if the chemical fingerprint of a birth cluster survives dispersal into the field, then the field's stars can be assigned to birth sites, and the disc's star-formation history can be reconstructed from the field alone, without the kinematic luck that today's analyses depend on. This workbook measures where that programme stands at our precision, and the honest summary is:

- **Established here.** Spectra contain cluster information that element-ratio summaries do not fully capture (§14); that information is enough to improve membership lists well beyond kinematic selection in the specific case we tested (**Table 12**), and it improves a physical parameter as a result; and distances from member giants agree with the literature for the clusters where we can measure them.
- **Not established.** That the improvement generalises. One cluster, one pipeline, and a sample with duplicate rows and a dominant globular cluster. A referee will ask for the same experiment on the 25-cluster set with five independent membership definitions, and they will be right to.
- **Not attempted here.** Recovering a dispersed cluster from the field with no kinematic prior. **Table 7** is our best attempt and it reaches precision ≈ 0.62 at recall 0.42. Good enough to sharpen a membership list, not good enough to find a cluster from scratch. Strong chemical tagging remains an open problem in the sense that Casamiquela et al. (2021) define it: their best-case sample, with differential abundances and no field stars, recovers 9 of 31 clusters at the 40 % threshold and one at 70 %, and their clustering step recovers less than that on our stars (§13.6).

EXERCISES

1. Reproduce the two-survey fit for one of the other clusters in the combined sample. Report age, distance and their posterior widths, with and without the red-clump prior. How much of the age improvement comes from the main sequence and how much from the prior?
2. The age–metallicity degeneracy means the posterior stays broad even with perfect membership. Choose a cluster in the sample and quantify the degeneracy directly: plot the posterior in the age–metallicity plane and measure its correlation coefficient. What extra observable would break it?
3. Contamination bias has a sign. Construct a synthetic cluster with 10 young field stars added, fit it with the same machinery, and compare the recovered age with the truth. Repeat with old contaminants. Which is the larger effect, and why?
4. **Table 12** removes 22 stars from the kinematic list and improves the age. Inspect those 22 stars: what are their abundances, their kinematics and their position relative to M 67? Write the three-sentence case for and against calling them contaminants.

16. THE ASSIGNMENT

Deck: slide 50, the hands-on brief.

This chapter is the brief. Everything before it is the material you need to execute it; everything you need to execute it is in this repository.

16.1. What you are producing

One object each (your assigned cluster) and one short report. The report makes exactly three claims, each with evidence:

1. **A membership list** for your cluster, with the criterion stated precisely enough that another student can reproduce it: the cuts, the parameters, the random seed, the commit hash.
2. **An honest score** for that list, against both available referees (kinematic selection and Simbad memberships), reported as recall *and* precision, and with the degenerate cases printed (§9).
3. **One physical parameter** derived from the list (an isochrone age and distance, or a red-clump distance) with its uncertainty, and a sentence on how the answer would change if your membership were 20% larger or smaller.

A report that claims a method is “better” without stating the population, the task, the seed count and both halves of the metric pair has not met the brief. This is the whole content of §9, applied to you.

16.2. Setup

The environment is reproducible from the repository root:

```
uv sync                # python + dependencies
uv run cluster --help  # the CLI
uv run cluster download # APOGEE DR19 + GALAH DR4 + Gaia DR3 subsets
```

Field runs are the slow ones. For interactive work, set `CLUSTER_FAST=1` to cap the field sample; every number quoted in this workbook was produced with `CLUSTER_FAST=0`. A container with the same environment is provided (`Dockerfile`), and the full walkthrough with worked examples is `docs/student_activities.md`.

16.3. Your cluster

Pair up on the open clusters; they are the hard part. Globulars tag cleanly already, which makes them a faster and solo-friendly track, and it is worth asking, while you work on one, why that is (§13: metallicity, age and the absence of recent star formation all push in the same direction).

16.4. The three tracks

Track A: spectral embeddings. Compare the 16 abundances with the 256-dimensional spectral latent on your cluster:

```
uv run cluster run --spectral data/embeddings/attention_broad_merged.parquet
```

cluster	kind	note	suggested track
M 3	globular	metal-poor (-1.6)	A (spectral wins)
M 5	globular		A
M 13	globular		A
M 15	globular	metal-poor (-2.2), southern	A or C
M 71	globular		A
M 107	globular	southern (DEC -13°)	C
M 92	globular	metal-poor (-2.3)	A
Pleiades	open	young (0.1 Gyr); chemistry fails here	A (expect a null)
King 7	open	sparse, faint	A
Berkeley 71	open	sparse	A
IC 166	open	sparse, contaminated	A
NGC 2158	open		A
NGC 1245	open		A
King 5	open		A
NGC 7789	open	giant-rich (1.4 Gyr)	B) clump
NGC 1798	open		A
NGC 2420	open		A
NGC 6819	open	giant-rich; clump residual $+0.01$	B
M 67	open	the workhorse; most data to beat	C (or A)
Berkeley 66	open	sparse	A
NGC 188	open	old (7.6 Gyr)	B
NGC 6791	open	old, metal-rich ($+0.4$)	B (or A)
Berkeley 17	open		A
NGC 2243	open	the sweet spot: main sequence <i>and</i> clump	B or C
Collinder 261	open	the sweet spot: 214 GALAH members	C

Table 13 The 25 clusters and the track each one suggests. Track assignments are advisory; the reasons are in the notes and in §13–§15. Metadata for every cluster, including the literature ages and distances you will be scored against, is in `src/cluster/literature.py`.

Best on the metal-poor globulars, where element ratios lose discriminating power and the spectrum retains it. Expect a null result on the Pleiades, and report it as a result.

Track B: isochrones and the red clump. Recover age and distance:

```
uv run python scripts/red_clump.py
uv run python scripts/sweet_spot.py
```

Best on the giant-rich open clusters (NGC 6819, NGC 7789). Compare against `src/cluster/literature.py` and be explicit about the age–metallicity degeneracy of §15.1.

Track C: two surveys. GALAH main sequence plus APOGEE giants:

```
uv run python scripts/build_galah_apogee.py
uv run python scripts/rerun_combined.py
```

Southern clusters only (DEC $\lesssim +25^\circ$). This is the track with the largest payoff and the largest number of ways to go quietly wrong: the parallax relaxation is required, not optional, and the two surveys’ abundance scales must be reconciled before the combined fit means anything.

16.5. Levers you may turn

One axis at a time, and log the configuration each time: the sweep in **Table 9** was only interpretable because every run recorded its settings.

lever	where	what it changes
TSNE.perplexity	src/cluster/config.py	neighbourhood size of the embedding
UMAP.n_neighbors	src/cluster/config.py	local graph scale
HDBSCAN.min_cluster_size	src/cluster/config.py	the size floor (removed by EVoC)
SNR_MIN	environment	sample quality
NORMALIZE_ROWS	environment	Euclidean $\leftrightarrow$ cosine equivalence
USE_ELEMENT_WEIGHTS	environment	element weighting, post-standardisation
DWARF_ONLY	environment	giant/dwarf selection

Table 14 The levers, and the one rule that applies to all of them: change one at a time, and report the effect with both halves of the metric pair. The test-suite defaults are in `src/cluster/config.py`; the measured effects on the DR17 sample are in `docs/experiment_results.md`, and they do not all survive on the current data.

16.6. How you are assessed

THE RUBRIC

Reproducibility (40%). A second person, on a clean checkout, gets your numbers. Configuration, seed, commit hash and the exact command are in your report. If you used a subsample, say how it was drawn.

Empirical care (40%). Both referees. Both halves of recall and precision. Mean $\pm$ s.d. over at least three seeds for any stochastic step. The largest-group fraction printed. A null result reported as a null result: a cluster that chemistry cannot separate is a finding, not a failure.

Physics (20%). One parameter with an uncertainty, a comparison with the literature value, and an honest statement of the dominant systematic, which in most cases will be your membership list, not your photometry.

EXERCISES

1. Write the one-paragraph methods section of your report before you run anything: the population, the cuts, the method, the seed, the metrics, the referees. If you cannot write it, you are not ready to run.
2. Take a published tagging result in the field (a table of members, or a claimed purity) and try to score it with this workbook's protocol. What information is missing from the paper, and what would you have asked the authors for?
3. Your cluster is listed with a suggested track. Write two sentences arguing for the other track, using only the cluster's properties (age, metallicity, declination, richness).

17. CONCLUSIONS

Deck: slides 48–49: the eight lessons and the take-away.

This chapter collects the eight lessons the benchmark produced, each with the number that supports it and the section that demonstrates the mechanism. They are the same eight the lecture closes on; here they are expanded to the length the evidence allows.

17.1. Eight lessons from the benchmark

1 · All-sky collapses; cut a region. One embedding of every clean star in the survey buries each cluster in the field: the field is 357 056 stars of chemically smooth background, and 25 clusters of a few dozen members each are not density peaks in that space. The published answer, and the one that works, is to restrict the sky. Kos et al. (2017) used 40° around the Pleiades, and our region runs use a cone scaled to the cluster, $\max(3^\circ, 10 \times \text{diameter})$. The cost of ignoring this is measurable: in the uniform all-sky run the abundance-only t-SNE arm reaches recall 0.201 at precision 0.240 (**Table 7**), while the same pipeline in region mode on the DR17 data reached recall 0.73 at precision 0.47 (**Table 8**). Same methods, same clusters; the difference is the contrast between the cluster and its immediate neighbourhood.

2 · Precision is the hard part. Read across the region-by-region sweep and the macro precision is 0.28 for t-SNE, 0.08 for UMAP and 0.01 for EVoC (25 clusters, DR19). The field is chemically similar to the clusters (that is what it means for a cluster to be dissolved in it) so a method that recovers members also recovers their look-alikes. We can now put a number on the mechanism that Casamiquela et al. (2021) report: they find that more than 70 % of the groups in their best-case solution are statistical mixtures, and our own cluster-only runs give 64–72 % of groups that recover no cluster at all (**stat40** in §13.6). Their negative result is about cluster-to-cluster separation and ours about field contamination, but both say the same thing: at survey precision, the chemical differences between clusters of the same age are comparable to the differences between a cluster and the field.

3 · Recall ≈ 1 is usually a blob. HDBSCAN* on the abundance space swallows the dense field into a single cluster, after which any best-overlap scorer reports that the true members are inside it: recall 1.000 at precision 0.0013 (UMAP on abundances, **Table 7**, and note that this run has row-normalisation *on* (it is the configuration default, and it does not prevent the collapse). Two defences are built into this workbook: report kNN purity, which cannot be gamed by a group’s size; and never quote recall without its precision. The third defence we tried) row-normalisation. Is the subject of the quiet lesson below.

4 · Globulars tag cleanly; open clusters do not. The kNN purity of the metal-poor globulars reaches 0.4–0.5 while solar-metallicity open clusters stay below 0.2. Three effects push the same way: globulars are older, more metal-poor, and formed from gas that no longer resembles the present-day interstellar medium, so their chemistry is distinct from the field they sit in. The Pleiades are the limiting case (100 Myr old, formed from gas the field still resembles) and the prediction that chemistry fails there is part of the assignment (§16), to be reported as a null result when it does.

5 · t-SNE isolates, UMAP over-merges, EVoC is fast but coarse. Three sets of assumptions, three answers, and no winner. t-SNE produces the most separated picture and is the least stable under row order (0.218–0.560 on the abundance matrix, §9). UMAP is the fastest and the only arm we run on the full field, and it is the arm that produces

blobs. EVoC needs no clustering choice from the user, works on the cosine geometry the data actually have, and has a seed spread (± 0.043 on abundances) that overlaps the gaps being claimed between methods.

6 • Levers are fragile, and read them in pairs. On the DR17 data, weighting each element by $1/\sigma$ *after* standardising lifted M 67’s t-SNE recall from 0.09 to 0.44: the largest single effect in the sweep, and the fix for a bug that had made the same switch a no-op. On the DR19 data the same switch moves recall from 0.10 to 0.96 while precision falls from 0.14 to 0.05: t-SNE’s cluster grows until it swallows the field. The region sweep agrees, with precision falling $0.83 \rightarrow 0.24$. A lever selected on one metric, one cluster and one seed is a hypothesis; it becomes a result only when it is re-run on the current data with both halves of the metric pair reported.

7 • Spectra beat abundances, and not PCA. A self-supervised masked autoencoder on the raw spectrum separates the 25 clusters better than the 16 ASPCAP abundances (0.760 vs 0.578 homogeneity, UMAP, the same 982 stars), and a plain PCA of the same spectra does about as well (0.754). Two very different representations of the same pixels land on the same score, which tells you where the limit is: not in the model, but in what the spectra and the metric jointly permit. On field retrieval the two tie to the third decimal (0.418/0.619 against 0.417/0.631), and the only field-retrieval claim we made for the spectral latent has been withdrawn for the reason given in §14.5.

8 • Two surveys, two parameters. APOGEE sees giants: the red clump gives distances, and NGC 6819 comes out within 0.01 mag of the literature value (**Table 11**). GALAH sees the main sequence and turnoff: that is where age sensitivity lives. The joint fit is not there yet (the age posterior ($\sigma_{\log a} \approx 0.97$) is close to the width of the flat prior, so the age half of the promise needs a better likelihood before it shows anything) but the distance half works, and the NGC 2243 experiment (§15) shows what the combination can do when the distance is pinned and the main sequence is added: an age residual improving from 0.52 to 0.09.

17.2. The quiet lesson

We used to say that row-normalisation breaks the blob, with a precision gain of three- to fivefold. Measured across the 25 clusters of the DR19 region sweep, it does not: turning normalisation *off* raises UMAP’s mean precision by $1.7\times$ ($0.08 \rightarrow 0.14$) and cuts the number of blobs from twelve to five, while t-SNE is unchanged ($0.28 \rightarrow 0.26$) and EVoC moves by a few thousandths. On M 67 the effect runs the other way: with normalisation on, t-SNE precision is 0.83; off, it is 0.28. The lever is real, but its direction depends on the method and the cluster, which is exactly what a lever selected on one cluster at one seed looks like when it is re-measured properly (§9). What survives as a rule is smaller and duller: the precision–recall trade is there in every setting, and it is a choice about which error you can live with, made explicit.

SUMMARY POINTS

1. Clustering is a decision problem: which space, which metric, which notion of “cluster”, which scale, and every choice belongs in the write-up.
2. Geometry beats algorithm. The largest improvements in this project came from changing the representation (spectra instead of abundances, row-normalisation, 64

components instead of 256), not from changing the clusterer.

3. Density-based clustering needs a hierarchy or a persistence criterion. One global threshold cannot serve a field of 357 056 stars and a cluster of 6 members at the same time.
4. Embeddings are for neighbourhoods, not distances. A t-SNE or UMAP map is evidence about who is next to whom, and about nothing else.
5. Validate with something the algorithm never saw, and print the degenerate cases. Largest-group fraction, number of groups, both halves of the metric pair.
6. Measure the ceiling and report the gap. Kinematics are not a competitor; they are the ceiling that chemistry is being compared with.
7. A lever is a hypothesis. Re-run it on the current data, with both halves of the metric pair, before quoting it.
8. Membership quality propagates into physics: M 67's isochrone age moves from 1.89 to 2.65 Gyr, against a literature value of 2.82, as the membership list improves.

FUTURE ISSUES

1. **Re-embed on a single data product.** The mixed-product latent of §14.5 invalidates every field-retrieval number computed from it. Uniform `mwmStar` spectra for the backfilled members are already downloaded (736 spectra), and the re-embedding run awaits the model checkpoint. Until it is done, the field-retrieval rows for the spectral latent should be read as provisional.
2. **Collapse the duplicate rows.** 298 member rows share an `APOGEE_ID` with another row. Until the sample is deduplicated, every macro average over clusters is weighted by something that is not a star.
3. **Benchmark against Spina et al. (2025).** §13.6 covers the published negative result (their clustering step on our stars, their metric triple on our scores) but a common-population comparison with the graph-attention autoencoder, on the same clusters and the same two tasks, is still the experiment this work is missing.
4. **Test the strong-tagging claim.** Everything here is measured with a kinematic prior available. Recovering a dispersed cluster from the field alone remains the open problem, and the honest current status is precision ≈ 0.62 at recall 0.42.
5. **A metric for high-dimensional chemical space.** The dilution result of §13 is a statement about Euclidean distance, not about chemistry. A metric that respects the compositional structure of abundance vectors might make concatenation behave, and nobody has demonstrated one.

DISCLOSURE AND PROVENANCE

Every number in this workbook is produced by code in the accompanying repository at a recorded commit, and the scripts that generate the tables are named in the sections that use them. Three things are worth distinguishing. (1) The duplicate-row and row-order audit of §9, the product-mismatch diagnosis of §14.5, and the withdrawal of the

field-retrieval claim for the mixed latent are the repository’s own findings, reported here as they stand. (2) The K-means row of **Table 4** was added for this workbook (`article/scripts/kmeans_baseline.py`) and had not been measured before. (3) The figures are the lecture’s, reused from the deck so that the workbook and the slides describe the same drawings; the animations are flattened to filmstrips, and the deck itself is the primary reference for anything that moves.

17.3. Acknowledgments

The workbook accompanies the 2026 IAA–SO school on advanced neural networks, and is built on the accompanying repository’s benchmark. The data are the public work of the SDSS-V/APOGEE, GALAH and Gaia collaborations, and the methods are the work of the authors cited throughout. The teaching figures and the dataset design come from the lecture; the isochrone fits use the PARSEC grids through AStECA and `emcee`; the pipeline is written on `numpy`, `pandas`, `scikit-learn`, `astropy`, `matplotlib` and `umap-learn`, with EVOc for the production runs. Errors of interpretation that remain are ours, and the exercises exist so that the next pass is yours.



Figure 36

The school programme and the reference library. The deck itself, with its animations, is at the URL on the title page and in the QR code of **Figure 1**.

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